PALO ALTO NETWORKS (PAN-OS)

Complete Presentation Slides & Architecture Master Deck (11+ Hours Video Course)

Source: I-MEDITA 11+ Hours Training Course | **Format:** High-Resolution Presentation Slide Captures + In-Depth Engineering Notes | **Compiled for:** Nishu Saini (Innspark Solutions)

Course Title & Instructor Masterclass Introduction

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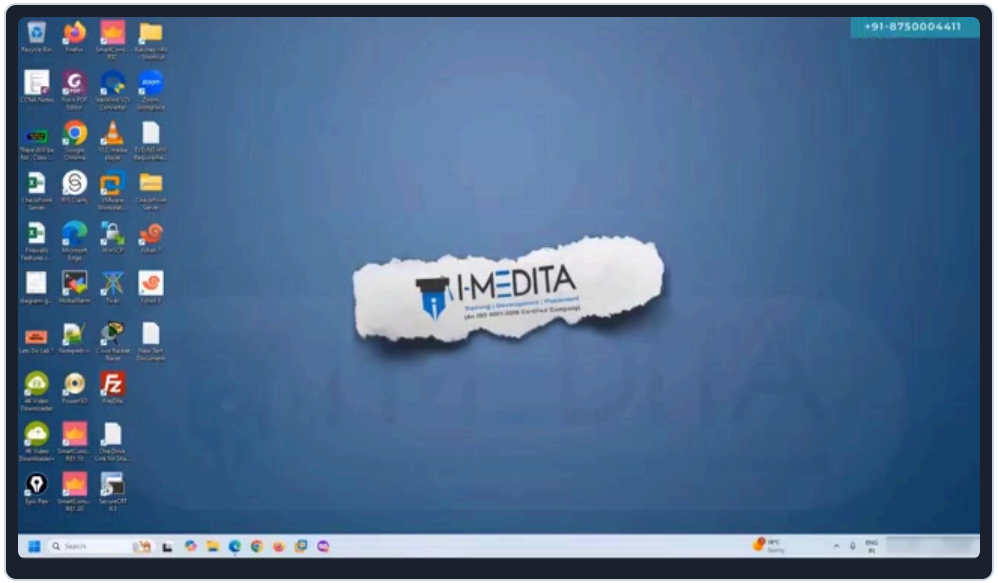


CORE CONCEPTS & KEY HIGHLIGHTS

- Comprehensive Palo Alto Networks Next-Generation Firewall Training (11+ Hours).
- PCNSE & PCCET certification alignment with real-world enterprise architectures.
- Instructor introduction and course expectations.

Complete Training Curriculum & Agenda Blueprint

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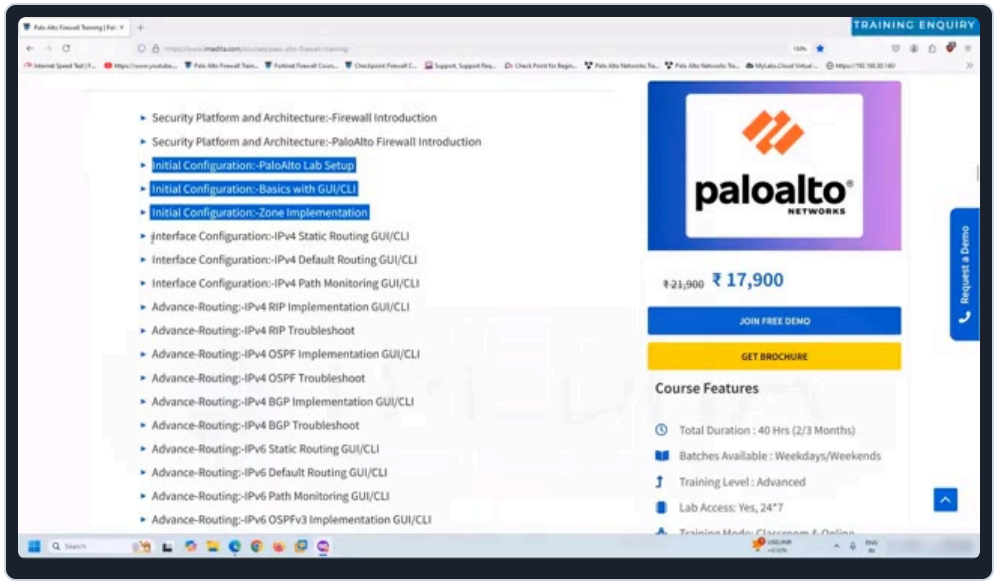


CORE CONCEPTS & KEY HIGHLIGHTS

- Roadmap covering Firewall Architecture, SP3, Hardware Engines, and Deployment Modes.
- Hands-on lab modules: Routing, NAT, Security Policies, App-ID, User-ID, Content-ID, VPN, and HA.
- Prerequisites and foundational networking concepts review.

Enterprise Lab Topology Blueprint & IP Planning

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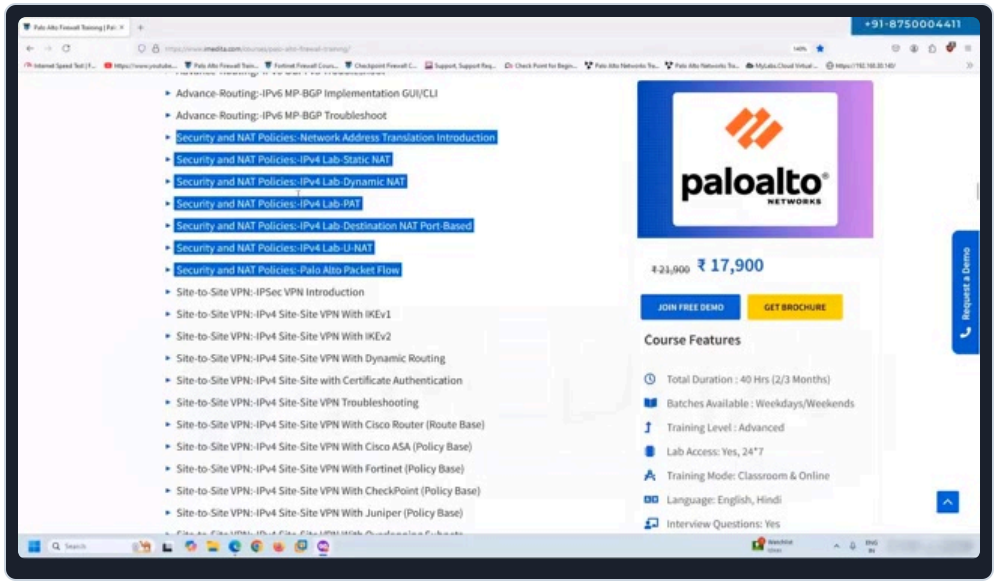


CORE CONCEPTS & KEY HIGHLIGHTS

- Network topology layout: Trust LAN (10.1.1.0/24), Untrust WAN (192.168.0.0/24), and MGT (192.168.1.0/24).
- Interconnecting PA-VM / PA-500 firewalls to Cisco core switches and edge routers.
- Setting up management access workstations for Web GUI and CLI console.

Enterprise Security Threat Landscape Evolution

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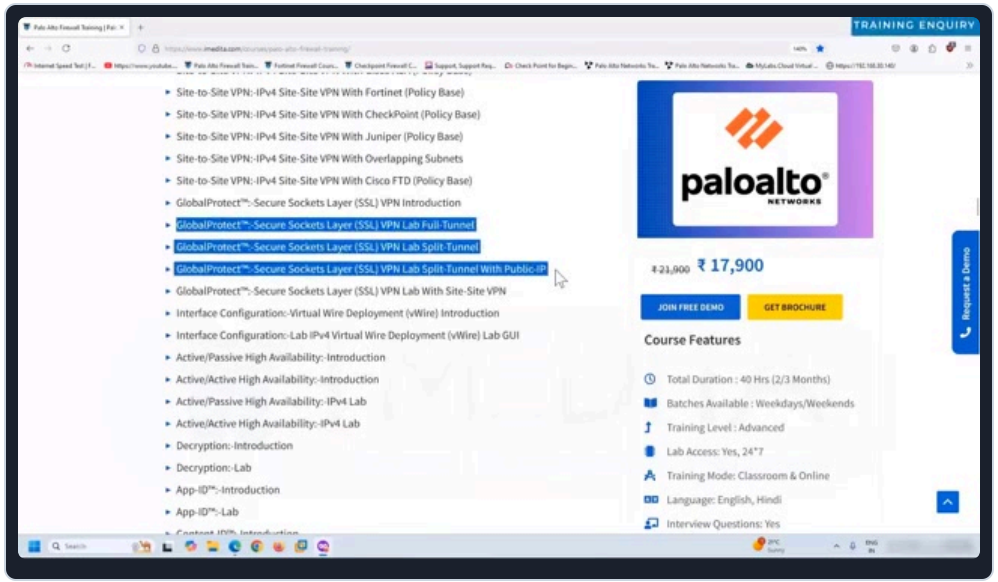


CORE CONCEPTS & KEY HIGHLIGHTS

- Evolution of threats: From simple port scans to sophisticated Layer 7 application exploits.
- Rise of ransomware, advanced persistent threats (APTs), and dynamic botnets.
- Why traditional network perimeter boundaries have dissolved in modern enterprises.

History & Evolution of Firewalls (Gen 1 to Gen 5)

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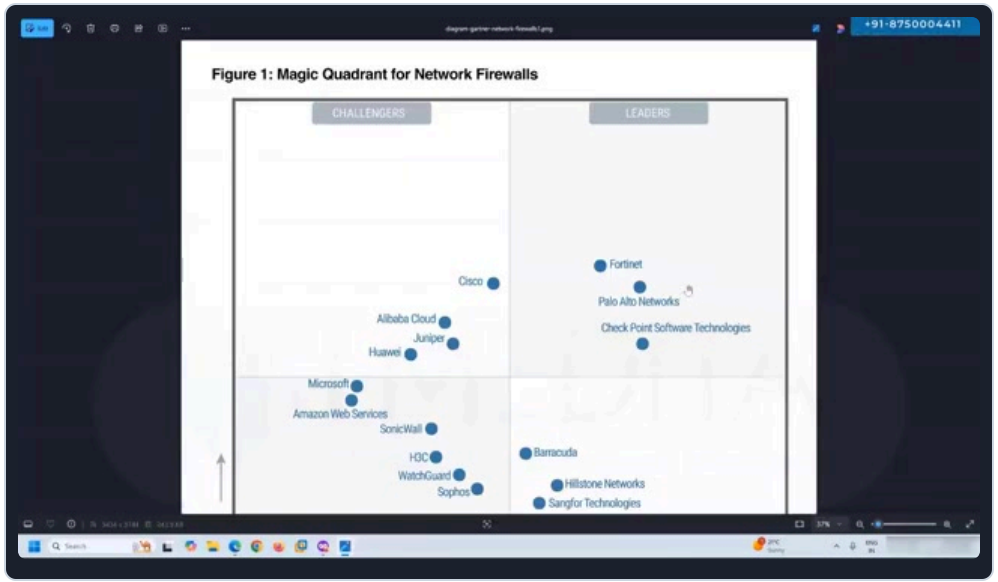


CORE CONCEPTS & KEY HIGHLIGHTS

- 1st Gen: Stateless Packet Filters (IP/Port ACLs without connection tracking).
- 2nd Gen: Stateful Inspection (TCP 3-way handshake and session table tracking).
- 3rd Gen: Application Layer Gateways & Proxies.
- 4th Gen: UTM (Unified Threat Management) multi-engine appliances.
- 5th Gen: Single-Pass Next-Generation Firewalls (NGFW).

Why Port-Based Traditional Firewalls Failed

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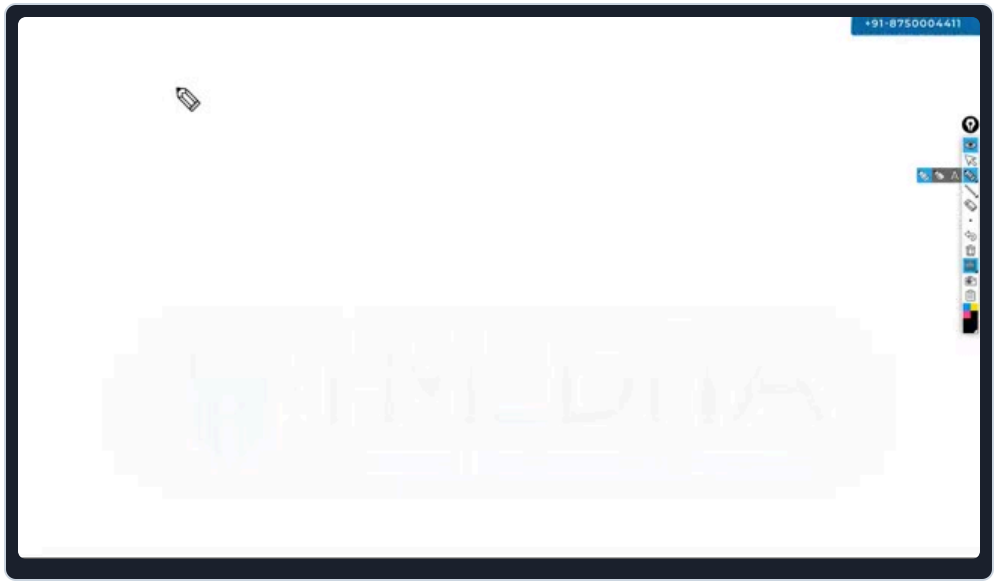


CORE CONCEPTS & KEY HIGHLIGHTS

- Modern applications do not use fixed, predictable port numbers.
- Dynamic port hopping protocols (BitTorrent, Skype, Zoom, RPC).
- Evasive protocols tunneling over open standard ports (TCP 80 HTTP, TCP 443 HTTPS).

Application Tunneling & Evasion Techniques

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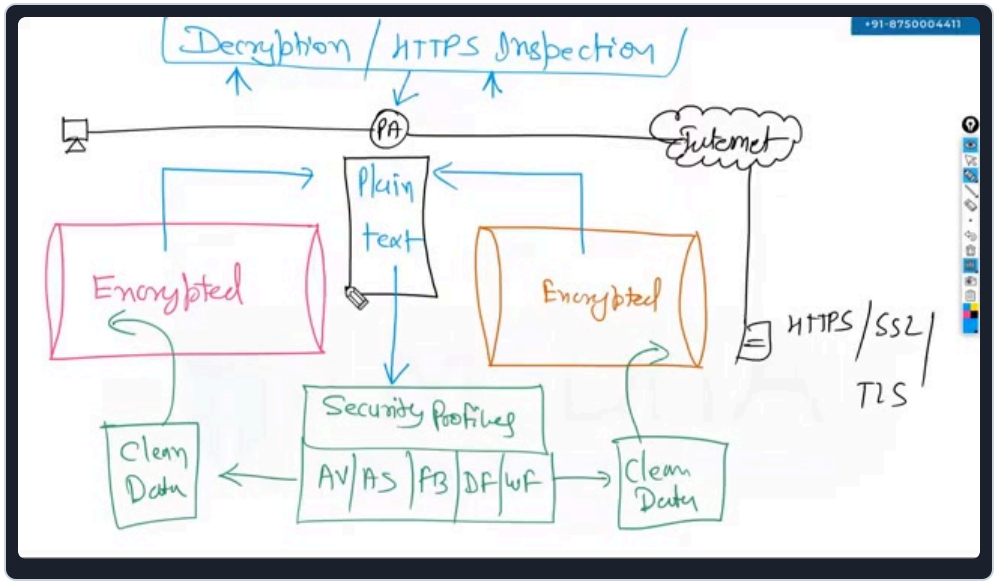


CORE CONCEPTS & KEY HIGHLIGHTS

- Non-web protocols wrapping inside HTTP/HTTPS payloads to bypass perimeter filters.
- Port hopping and SSL encryption shielding malicious payloads from traditional inspection.
- Need for Layer 7 payload decoders that classify traffic regardless of port.

Core Requirements for Next-Generation Firewalls

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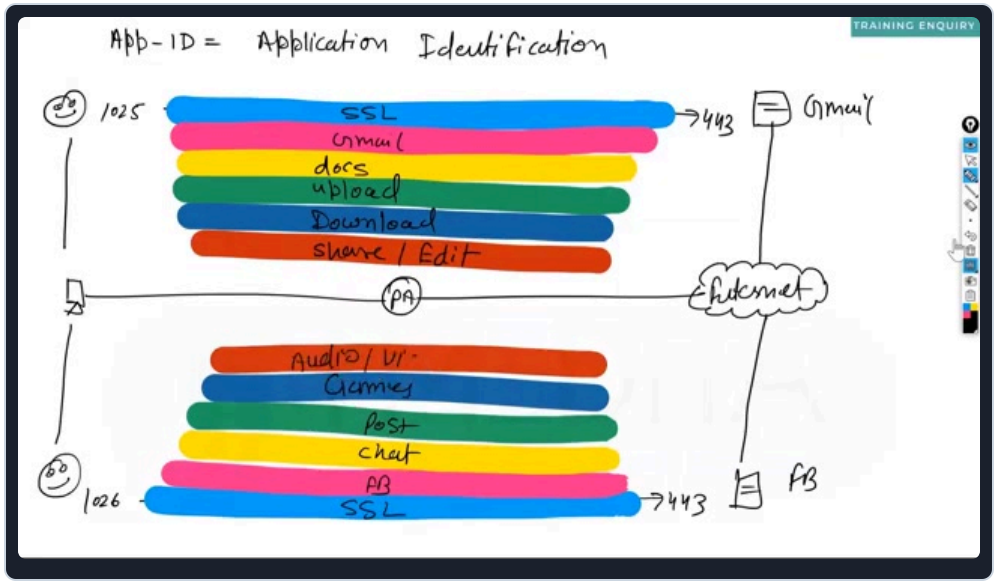


CORE CONCEPTS & KEY HIGHLIGHTS

- Application visibility and granular control across all ports at all times.
- Integrated real-time threat prevention (IPS, Antivirus, Anti-Spyware, URL Filtering).
- User and group identity-based policies instead of static IP addresses.
- High-performance line-rate throughput under full inspection.

Architectural Breakdown: UTM vs Palo Alto SP3

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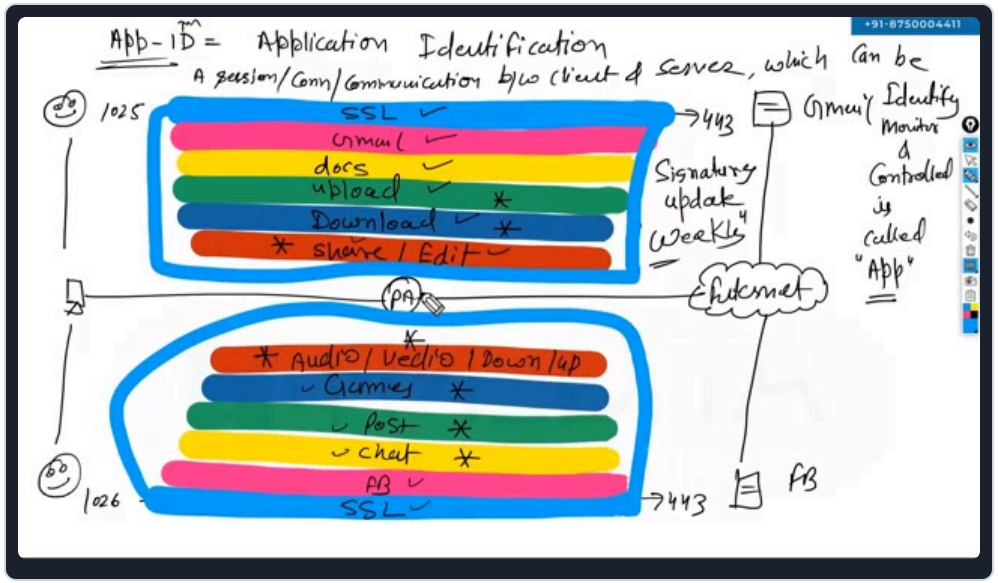


CORE CONCEPTS & KEY HIGHLIGHTS

- UTM Flaws: Sequential daisy-chaining of independent security engines (AV -> IPS -> URL).
- UTM Latency: Packets disassembled and reassembled multiple times causing throughput collapse.
- Palo Alto SP3: Single packet decode and single uniform signature matching scan.

SP3 Software Engine & Uniform Signature Scan

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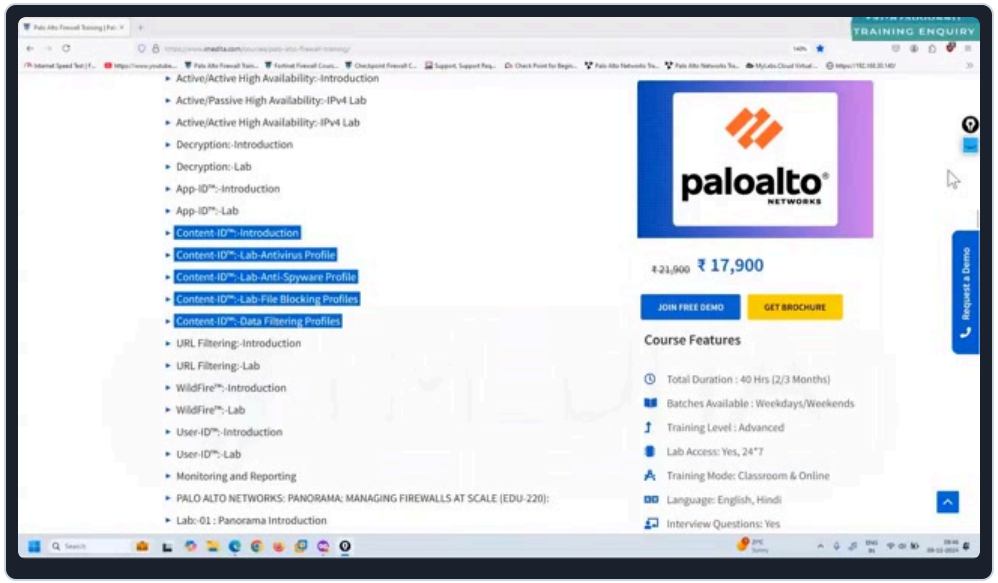


CORE CONCEPTS & KEY HIGHLIGHTS

- Uniform Signature Matching: All threat signatures compiled into one unified matrix.
- Simultaneous parallel execution of App-ID, User-ID, and Content-ID.
- Consistent, predictable performance with all security features enabled.

Hardware Processing Engines: NPE, SPE & SME

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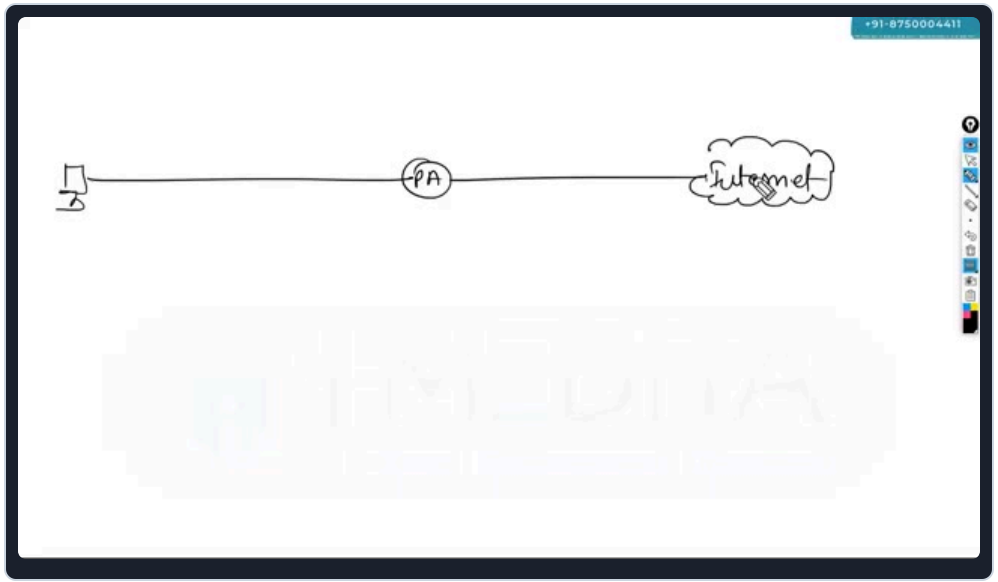


CORE CONCEPTS & KEY HIGHLIGHTS

- Network Processor (NPE): Wire-speed L2/L3 switching, routing, and NAT offload.
- Security Processor (SPE): Hardware cryptographic coprocessor for SSL/TLS and IPsec.
- Signature Matching Engine (SME): Custom Match Engine ASICs for parallel RegEx scanning.

Control Plane vs Data Plane Hardware Separation

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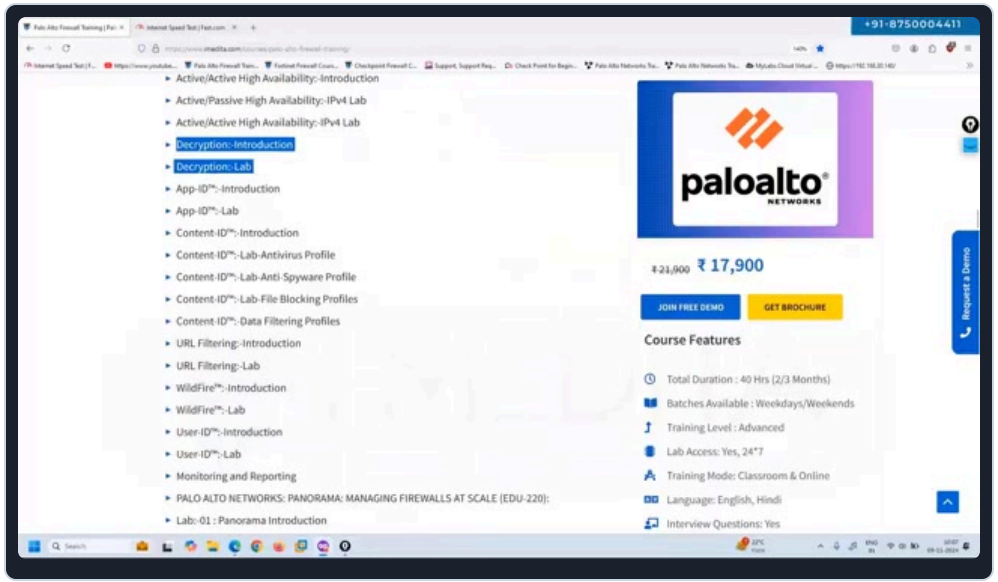


CORE CONCEPTS & KEY HIGHLIGHTS

- Management Plane: Dedicated CPU/RAM for Web GUI, CLI, Logging, and Panorama Sync.
- Data Plane: Dedicated multi-core CPU and ASICs for line-rate packet forwarding.
- Zero performance impact: Generating massive audit reports never slows network traffic.

The 4 Foundational Pillars of PAN-OS

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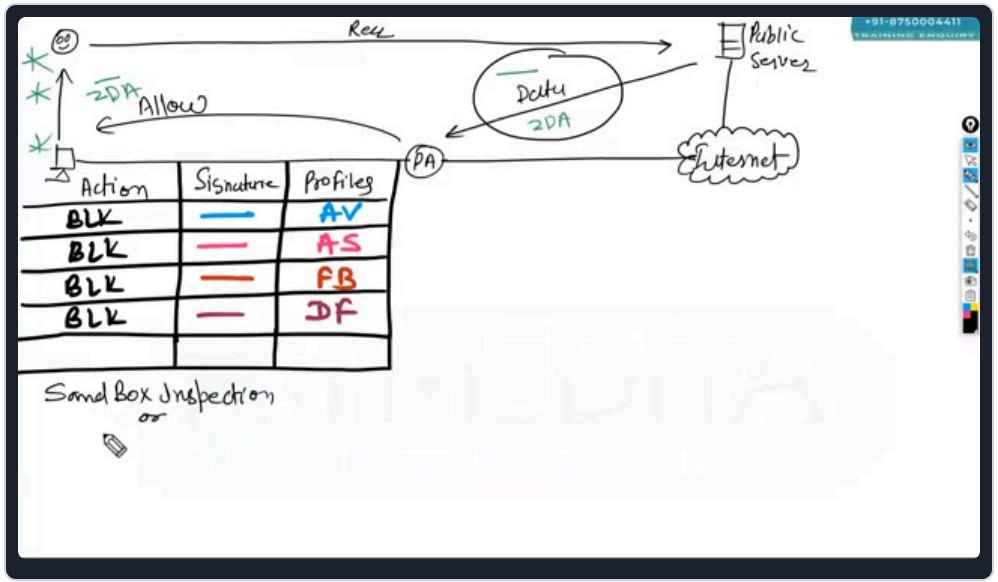


CORE CONCEPTS & KEY HIGHLIGHTS

- App-ID: Layer 7 application classification independent of port or protocol.
- User-ID: Active Directory and LDAP identity-based security policies.
- Content-ID: Integrated stream-based Antivirus, Anti-Spyware, and IPS.
- WildFire: Automated cloud-based zero-day malware detonation and signature generation.

Zero-Day Attacks & The Patient Zero Threat Model

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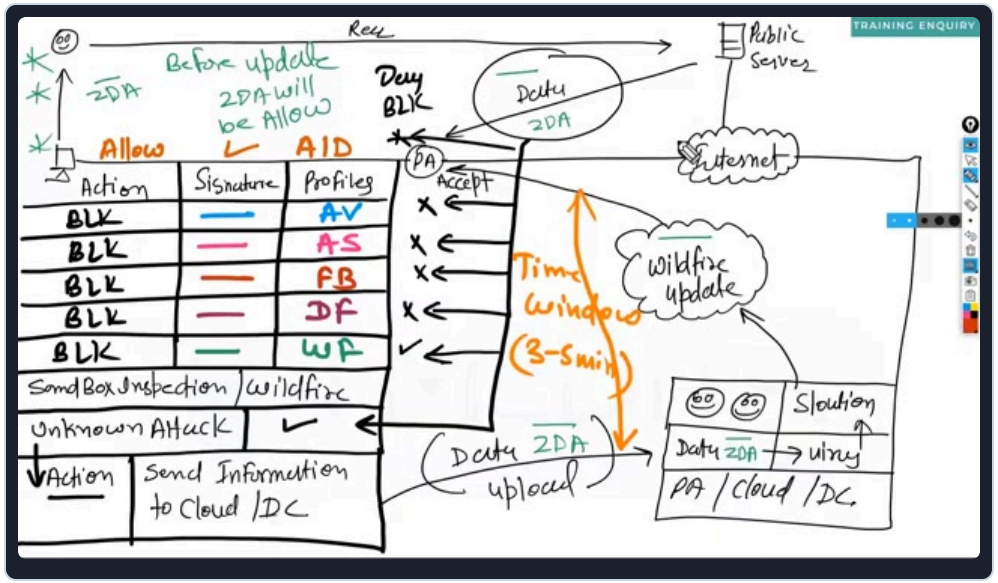


CORE CONCEPTS & KEY HIGHLIGHTS

- Why signature-based detection fails against newly authored malware variants.
- The concept of 'Patient Zero' and the window of vulnerability before signatures exist.
- Need for automated, dynamic behavioral sandbox analysis.

Palo Alto WildFire Sandbox Architecture

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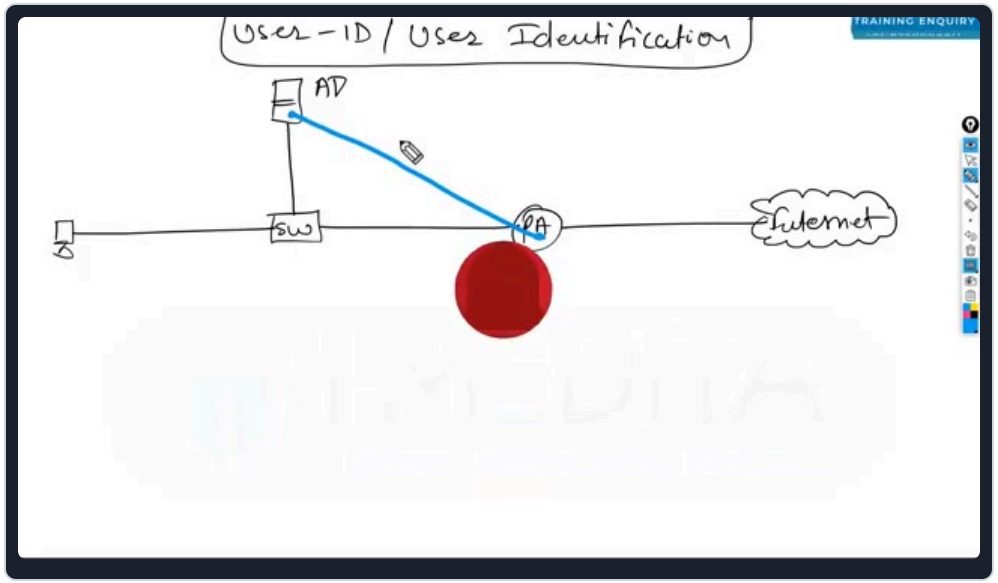


CORE CONCEPTS & KEY HIGHLIGHTS

- Automated extraction and forwarding of unknown executable files to WildFire cloud.
- Virtual multi-OS detonation observing behavioral anomalies (registry, C2 callbacks).
- Worldwide distribution of newly created threat signatures in under 5 minutes.

Stateless ACLs vs Dedicated Stateful Firewalls

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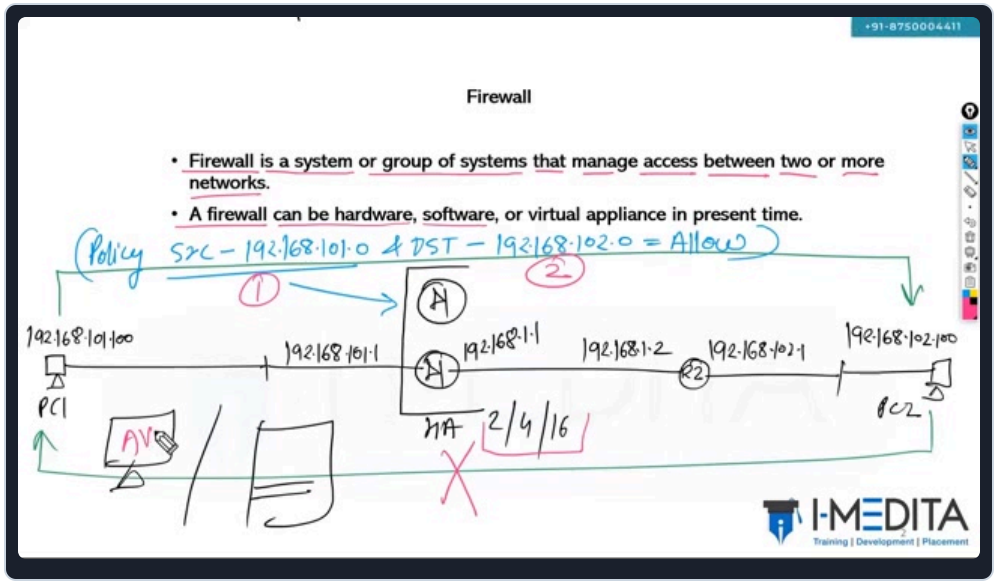


CORE CONCEPTS & KEY HIGHLIGHTS

- Stateless router ACLs require explicit bi-directional rules (inbound and outbound).
- Rule explosion, administrative overhead, and security holes in stateless filtering.
- Inability of stateless ACLs to track TCP sequence numbers or dynamic port negotiations.

Packet Filtering Mechanics & Header Inspection

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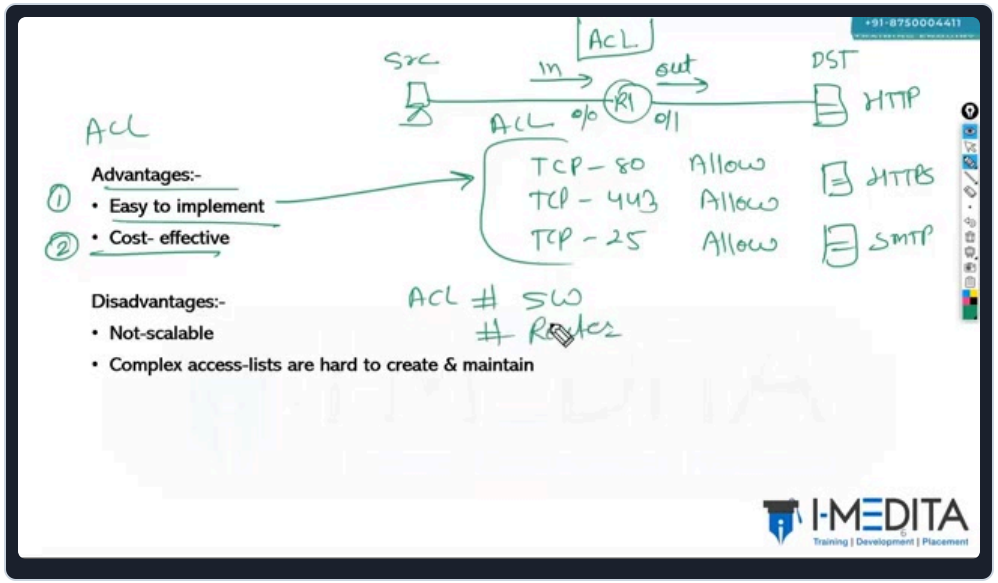


CORE CONCEPTS & KEY HIGHLIGHTS

- Inspection of Layer 3 IP headers (Source/Dest IP, Protocol) and Layer 4 TCP/UDP ports.
- Lack of payload inspection allowing malicious data in permitted ports.
- Why Layer 3/4 filtering is insufficient for modern application security.

Proxy Architectures: Explicit vs Transparent

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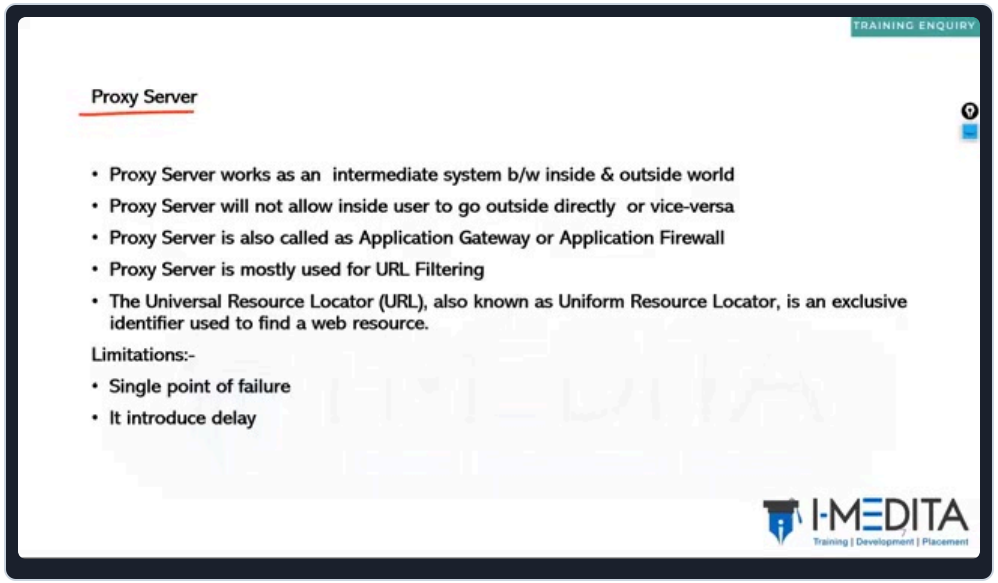


CORE CONCEPTS & KEY HIGHLIGHTS

- Dual-socket termination architecture breaking direct client-to-server TCP connections.
- Explicit Proxy: Client browser configuration (PAC/WPAD) redirecting traffic to proxy.
- Transparent Proxy: Network-level interception (WCCP/PBR) without client reconfiguration.
- Scaling limits: Heavy RAM consumption and protocol latency on traditional proxies.

Stateful Inspection & Session Table Architecture

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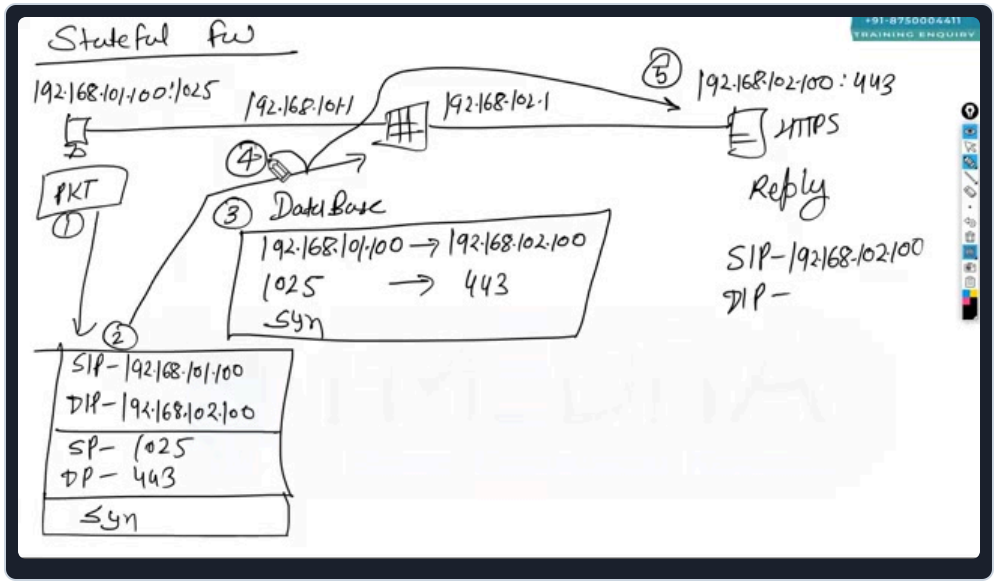


CORE CONCEPTS & KEY HIGHLIGHTS

- Tracking TCP 3-way handshake (SYN, SYN-ACK, ACK) and connection lifecycle.
- Session State Table: 5-Tuple (Src IP, Dst IP, Src Port, Dst Port, Protocol) + TCP State.
- Fast-Path forwarding: Dynamic return traffic automatically permitted without explicit inbound rule.

TCP State Machine Tracking & Anti-Spoofing

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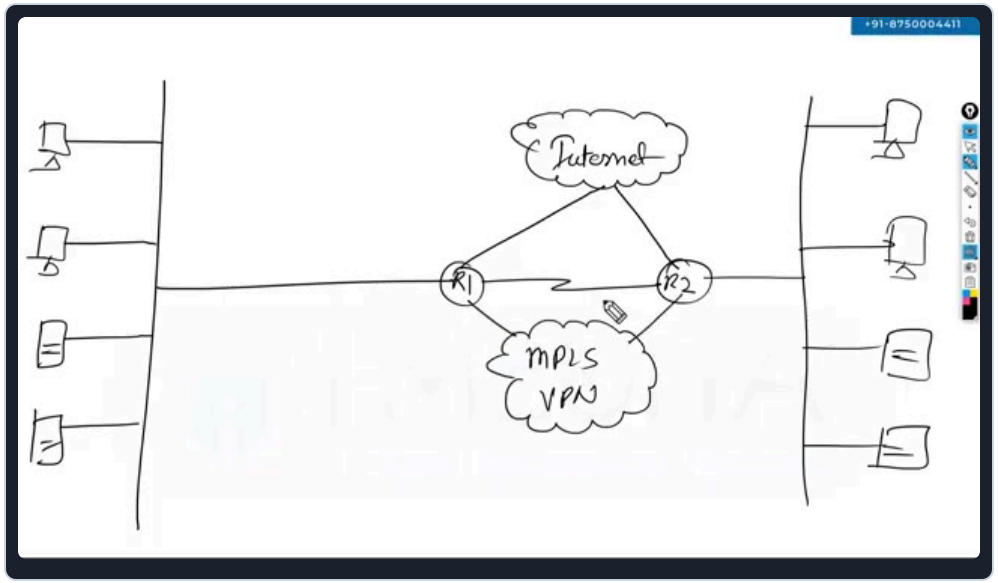


CORE CONCEPTS & KEY HIGHLIGHTS

- Validating TCP sequence and acknowledgment numbers to prevent session hijacking.
- Strict TCP window validation and teardown tracking (FIN / RST packets).
- Drop policies for out-of-state packets and malformed TCP segments.

Palo Alto Deployment Modes Overview

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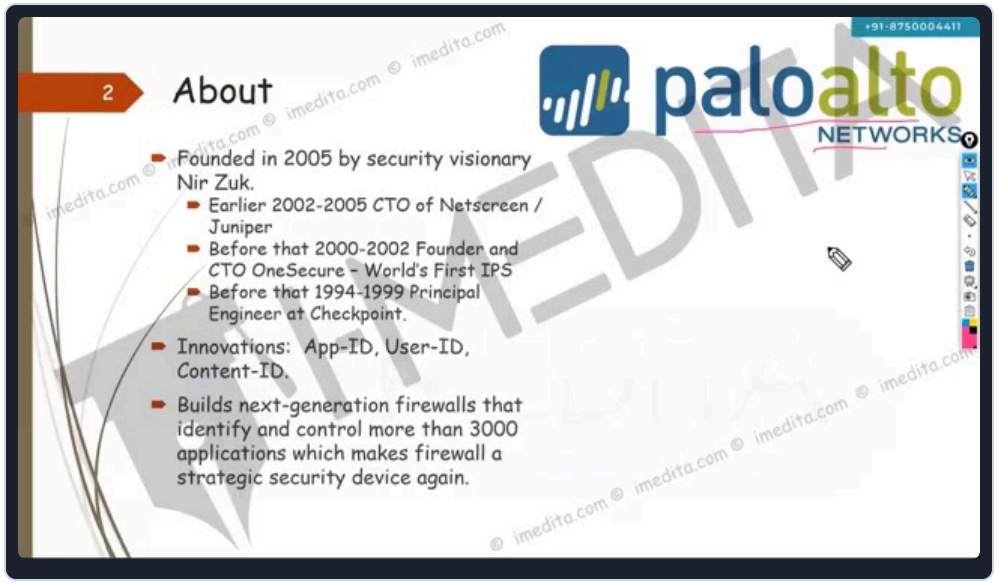


CORE CONCEPTS & KEY HIGHLIGHTS

- Layer 3 Routed Mode: Full routing, NAT, and gateway functionality.
- Layer 2 Mode: Bridging between VLANs with Next-Gen inspection.
- Virtual Wire (V-Wire): Transparent inline bump-in-the-wire without IP changes.
- TAP Mode: Passive SPAN/mirror port traffic monitoring and visibility.

Virtual Wire (V-Wire / Transparent Mode) Deep Dive

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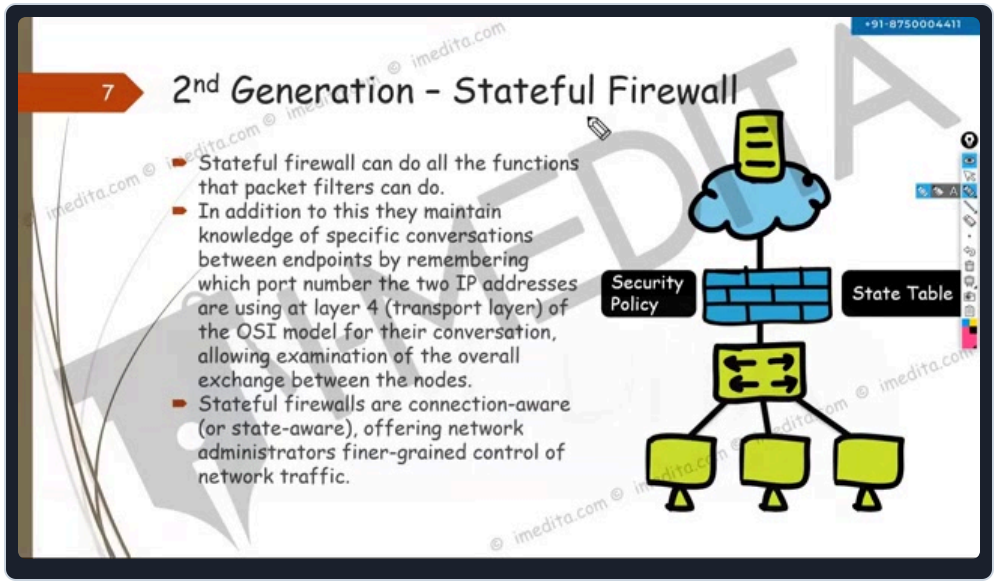


CORE CONCEPTS & KEY HIGHLIGHTS

- Zero network redesign: Interfaces have NO IP addresses and NO MAC addresses.
- Binds two physical ports into an inline transparent inspection tunnel.
- Ideal for immediate POC deployments and inserting security between existing core switches.

Firewall Generations Timeline & Nir Zuk History

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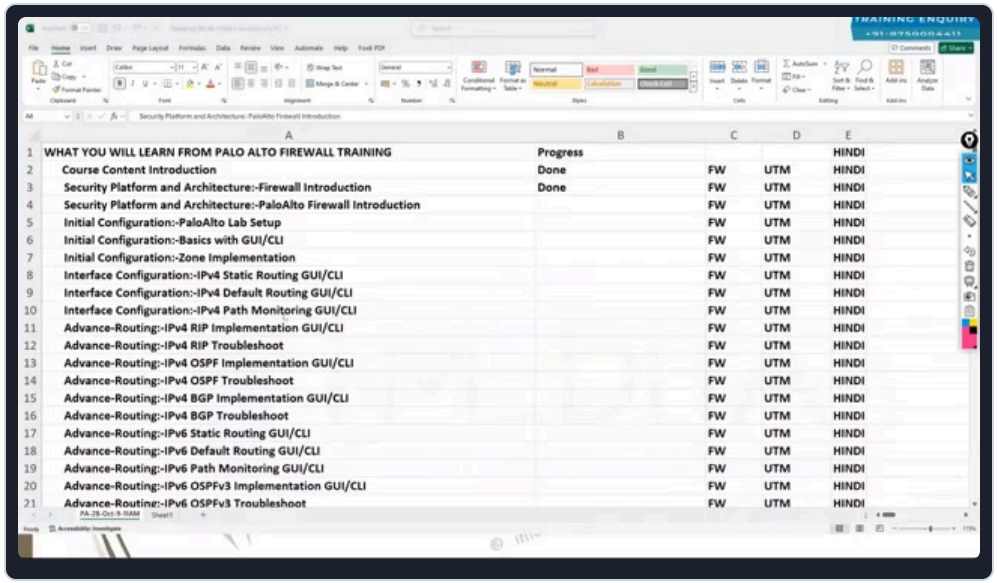


CORE CONCEPTS & KEY HIGHLIGHTS

- From Check Point FireWall-1 (1994) to Palo Alto Networks (2007+).
- Nir Zuk's pioneering innovations: Stateful Inspection -> Hardware IPS -> NGFW.
- How Palo Alto revolutionized the security industry with App-ID and SP3 architecture.

UTM Architecture Performance Bottlenecks

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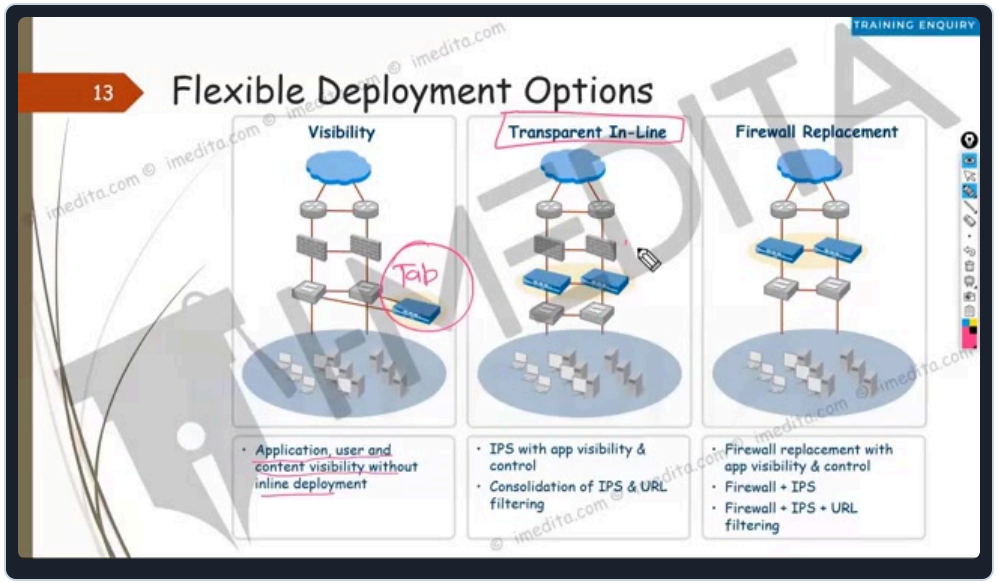
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3	Security Platform and Architecture:-Firewall Introduction	Done	FW	UTM	HINDI
4	Security Platform and Architecture:-PaloAlto Firewall Introduction		FW	UTM	HINDI
5	Initial Configuration:-PaloAlto Lab Setup		FW	UTM	HINDI
6	Initial Configuration:-Basics with GUI/CLI		FW	UTM	HINDI
7	Initial Configuration:-Zone Implementation		FW	UTM	HINDI
8	Interface Configuration:-IPv4 Static Routing GUI/CLI		FW	UTM	HINDI
9	Interface Configuration:-IPv4 Default Routing GUI/CLI		FW	UTM	HINDI
10	Interface Configuration:-IPv4 Path Monitoring GUI/CLI		FW	UTM	HINDI
11	Advance-Routing:-IPv4 RIP Implementation GUI/CLI		FW	UTM	HINDI
12	Advance-Routing:-IPv4 RIP Troubleshoot		FW	UTM	HINDI
13	Advance-Routing:-IPv4 OSPF Implementation GUI/CLI		FW	UTM	HINDI
14	Advance-Routing:-IPv4 OSPF Troubleshoot		FW	UTM	HINDI
15	Advance-Routing:-IPv4 BGP Implementation GUI/CLI		FW	UTM	HINDI
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18	Advance-Routing:-IPv6 Default Routing GUI/CLI		FW	UTM	HINDI
19	Advance-Routing:-IPv6 Path Monitoring GUI/CLI		FW	UTM	HINDI
20	Advance-Routing:-IPv6 OSPFv3 Implementation GUI/CLI		FW	UTM	HINDI
21	Advance-Routing:-IPv6 OSPFv3 Troubleshoot		FW	UTM	HINDI

CORE CONCEPTS & KEY HIGHLIGHTS

- Independent proxies and scanning daemons competing for system RAM and CPU.
- Multiple packet disassembly/reassembly stages creating significant packet latency.
- Why enabling UTM security features causes a 70-80% throughput drop on legacy firewalls.

Palo Alto SP3 Hardware Acceleration Advantages

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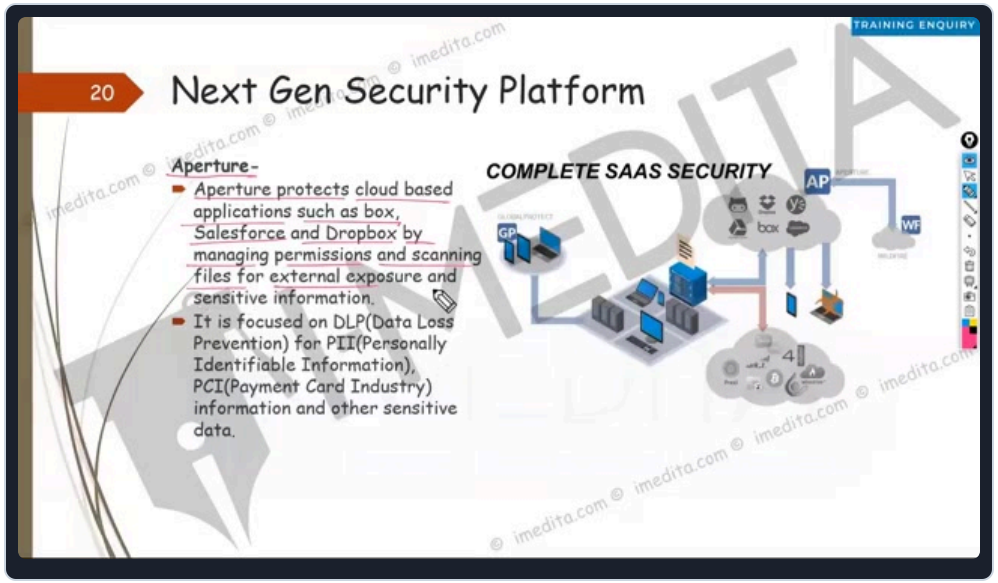


CORE CONCEPTS & KEY HIGHLIGHTS

- Predictable wire-speed throughput regardless of how many security features are enabled.
- Dedicated Control Plane ensures administrative tasks never degrade packet processing.
- Parallel hardware coprocessors dedicated to cryptographic offload and regex threat scans.

Palo Alto Hardware Appliance Portfolios

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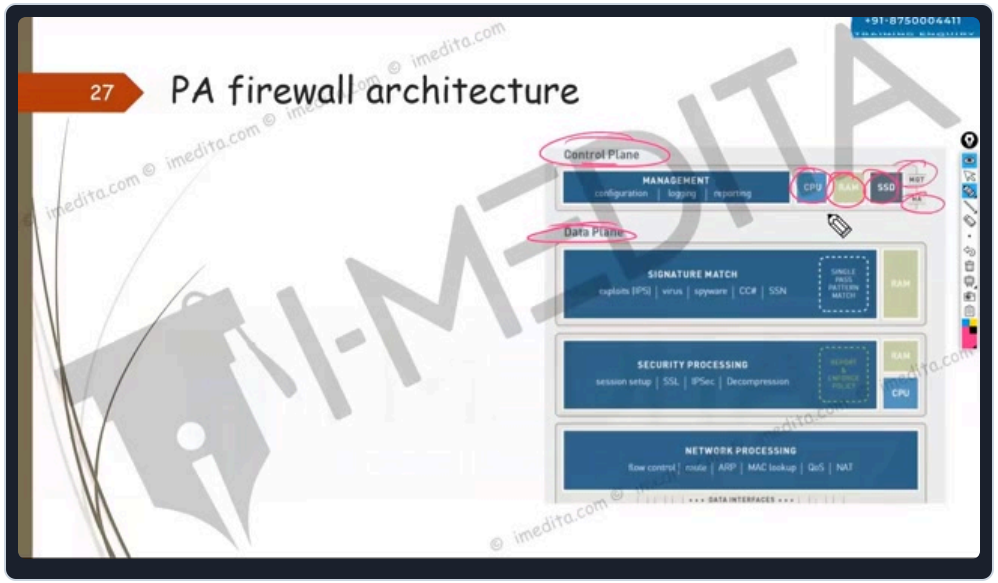


CORE CONCEPTS & KEY HIGHLIGHTS

- Branch Office: PA-220, PA-400 Series (PA-410, PA-440, PA-450, PA-460).
- Mid-Enterprise: PA-500, PA-800 Series (PA-820, PA-850), PA-3200 / PA-3400 Series.
- Data Center / Campus Core: PA-5200 Series (PA-5220/5250/5260), PA-5400 Series.
- Modular Chassis: PA-7000 Series (PA-7050, PA-7080) for mega data centers and ISPs.
- Virtual Firewalls: VM-Series (VM-50, VM-100, VM-300, VM-500, VM-700, VM-Flex).

Hardware Port Types & Interface Roles

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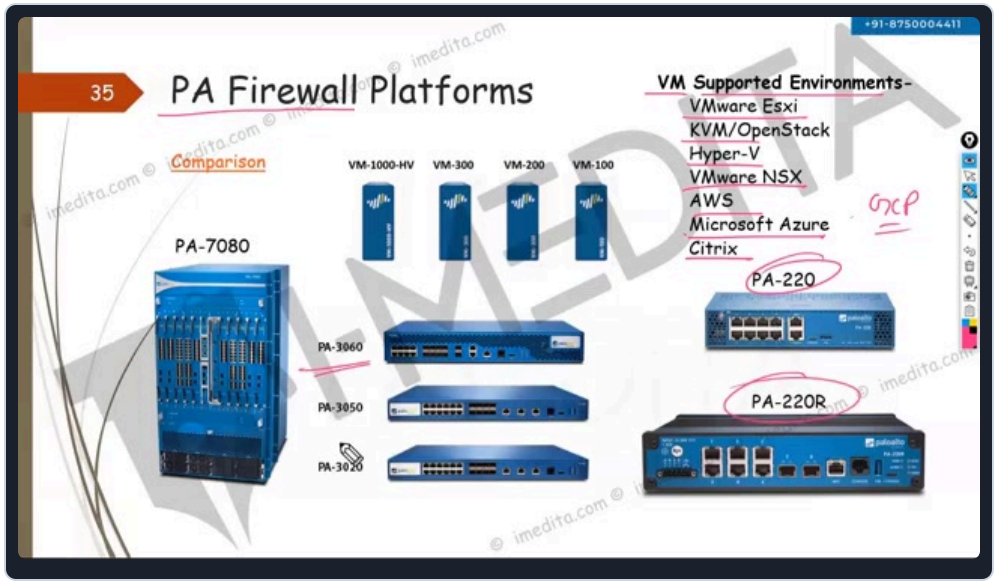


CORE CONCEPTS & KEY HIGHLIGHTS

- Management (MGT) Port: Out-of-band 10/100/1000 copper RJ-45 interface (Default IP: 192.168.1.1).
- Console Ports: RJ-45 Serial (9600-8-N-1) and Micro-USB for initial setup.
- High Availability Ports: Dedicated HA1 (Control/Heartbeat) and HA2 (Session Data Sync).
- HSCI Port: High-Speed Cluster Interconnect (10G/40G) on enterprise platforms.
- Data Plane Ports: 1G Copper, 1G/10G SFP+, 40G/100G QSFP28 interfaces.

Front Panel LED Status Diagnostics

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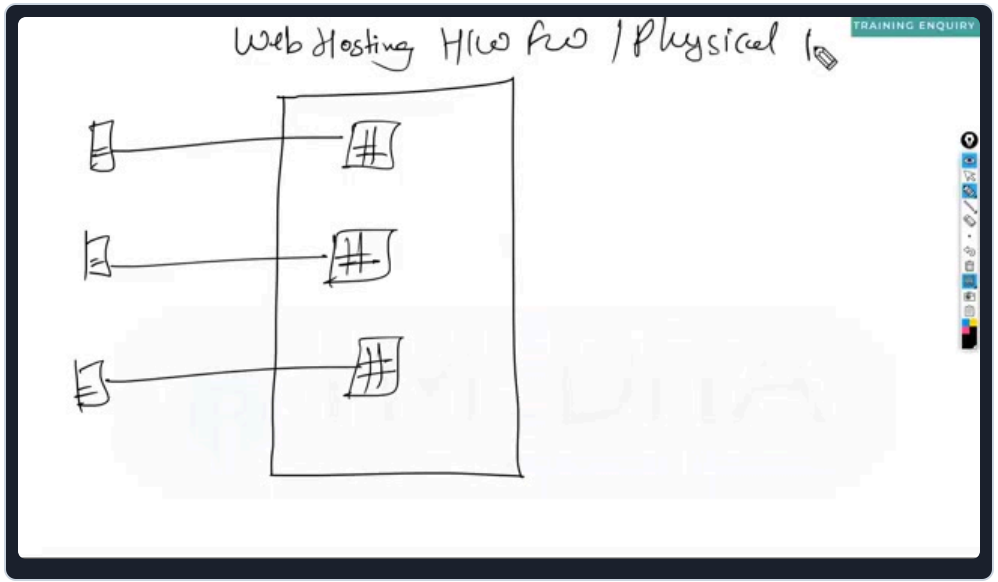


CORE CONCEPTS & KEY HIGHLIGHTS

- PWR LED: Green = Redundant power supplies normal; Amber = Power fault / single PSU down.
- STS LED: Green = PAN-OS operational; Amber = Booting / initializing; Red = System alarm.
- ALM LED: Off = Normal; Amber = Low severity alarm; Red = High severity hardware alarm.
- HA LED: Green = Active; Amber = Passive; Off = Standalone / HA disabled.

Console Connection & Serial Parameters

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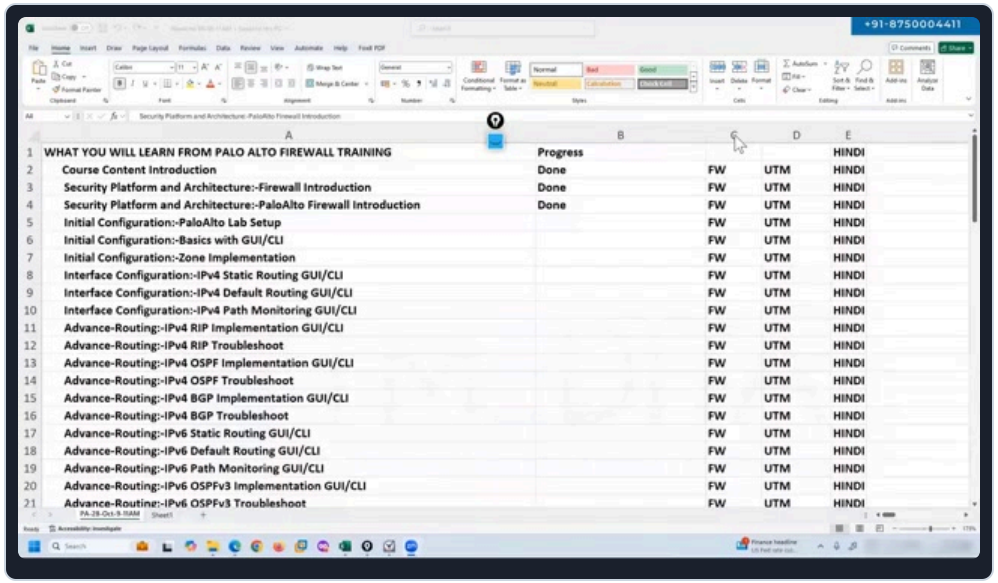


CORE CONCEPTS & KEY HIGHLIGHTS

- Connecting via RJ-45 to DB9 / USB-to-Serial rollover console cable.
- PuTTY / SecureCRT Terminal Settings: Baud Rate 9600, Data Bits 8, Parity None, Stop Bits 1, Flow Control None.
- Default administrative credentials: admin / admin (forced password change on initial login).

Hands-On Lab Topology Interconnect & Setup

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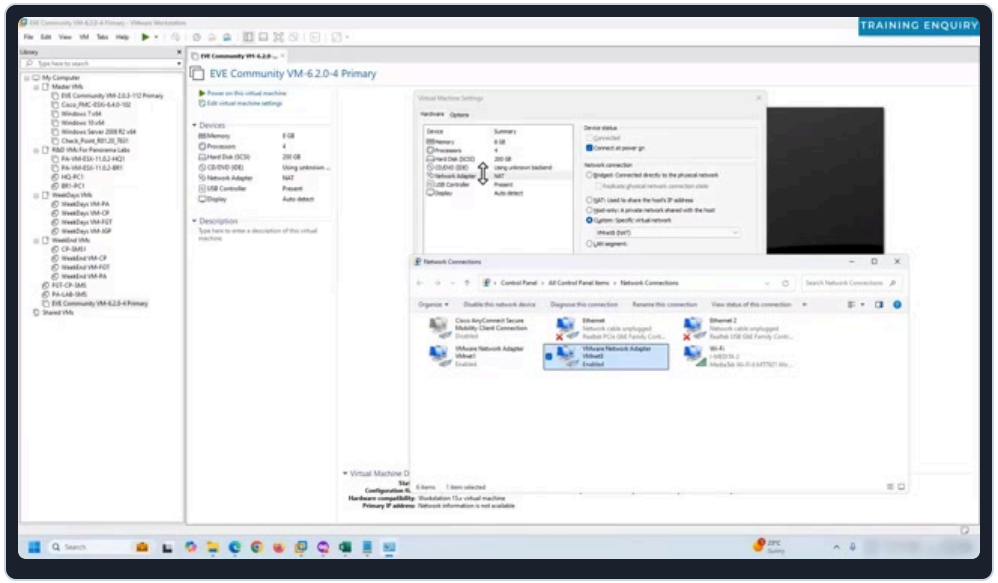


CORE CONCEPTS & KEY HIGHLIGHTS

- Connecting Admin Management PC to MGT port via HTTPS (https://192.168.1.1).
- Configuring ethernet1/1 as Trust LAN interface and ethernet1/2 as Untrust WAN interface.
- Interconnecting firewall to core switch and upstream internet gateway.

Hands-On Enterprise Lab Topology Design

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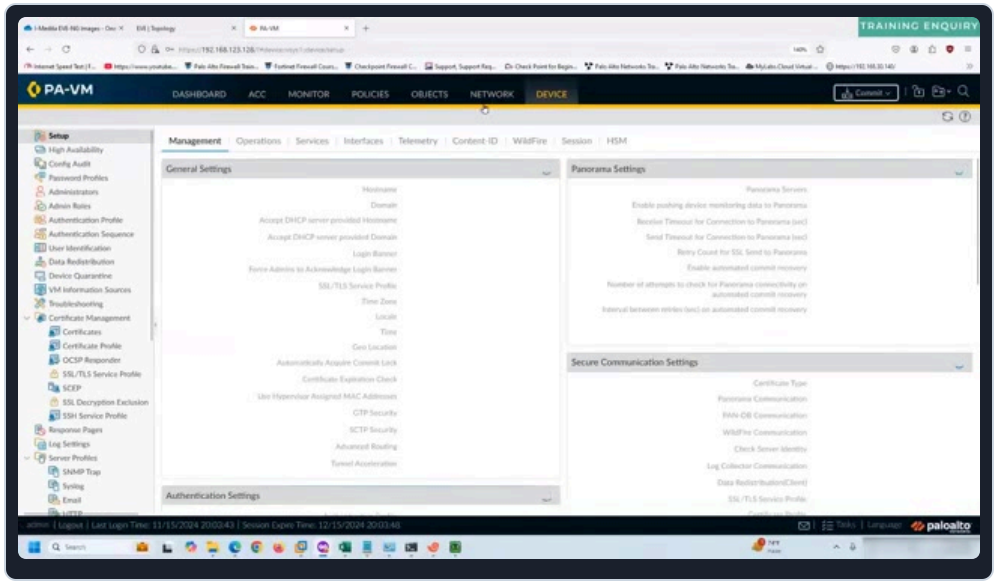


CORE CONCEPTS & KEY HIGHLIGHTS

- Architecture of Trust LAN (10.1.1.0/24), Untrust WAN (192.168.0.0/24), and MGT.
- Connecting admin PC via HTTPS/SSH to PA-VM-1 / PA-500 management port.
- Core switch and ISP router interconnection layout.

Initial PAN-OS Access & CLI Command Structure

Timestamp: 03:30:00

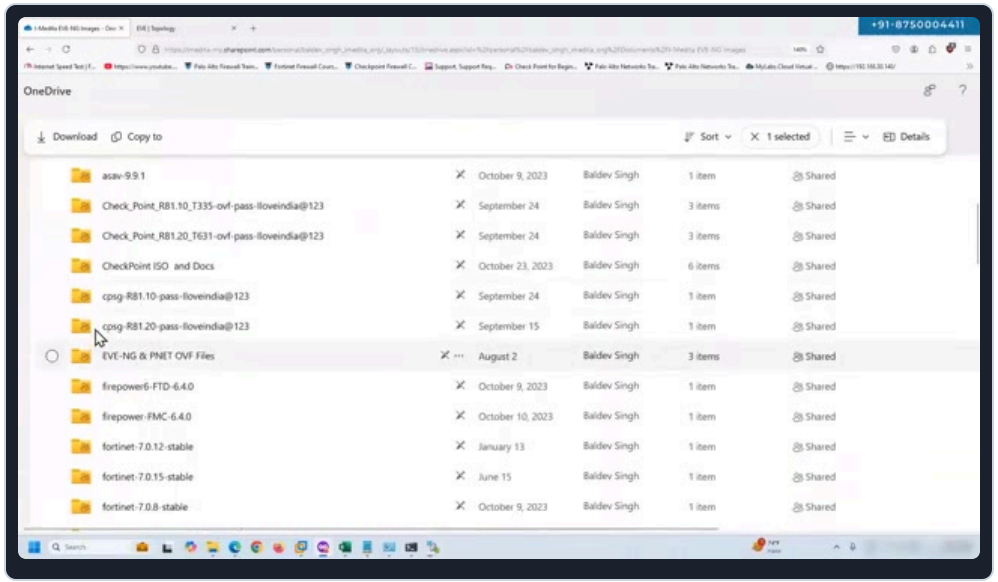


CORE CONCEPTS & KEY HIGHLIGHTS

- Default login credentials (admin/admin) and forced password change.
- Operational mode vs Configuration mode (`configure`) commands.
- `show`, `set`, `request`, and `test` command taxonomy.

Candidate Config, Running Config & Commit Engine

Timestamp: 03:50:00

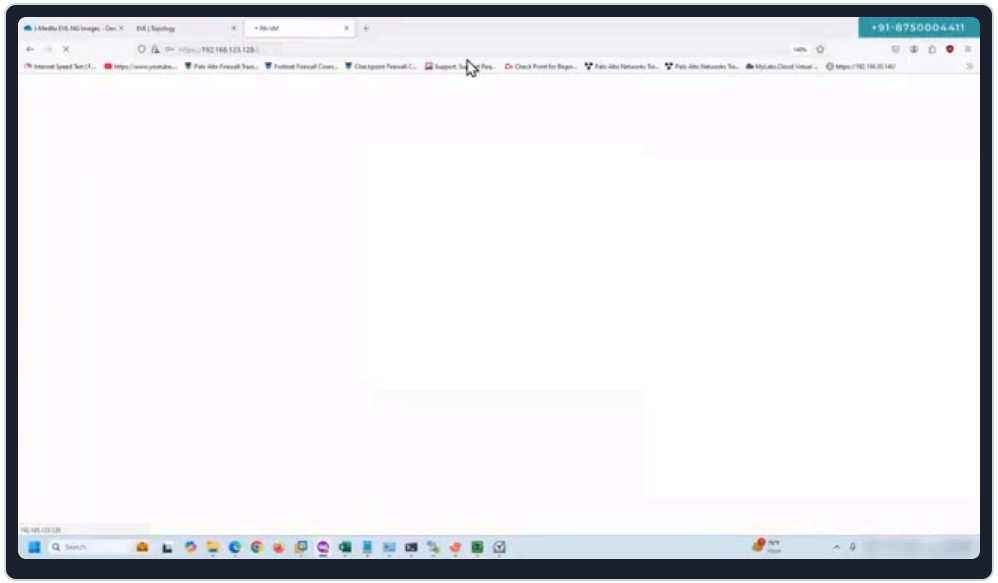


CORE CONCEPTS & KEY HIGHLIGHTS

- Candidate Configuration draft changes vs active Running Configuration.
- Commit operation compiling rules into hardware ASICs.
- Reverting uncommitted changes and utilizing multi-admin config locks.

Virtual Systems (VSYS) Fundamentals & Overview

Timestamp: 03:58:00

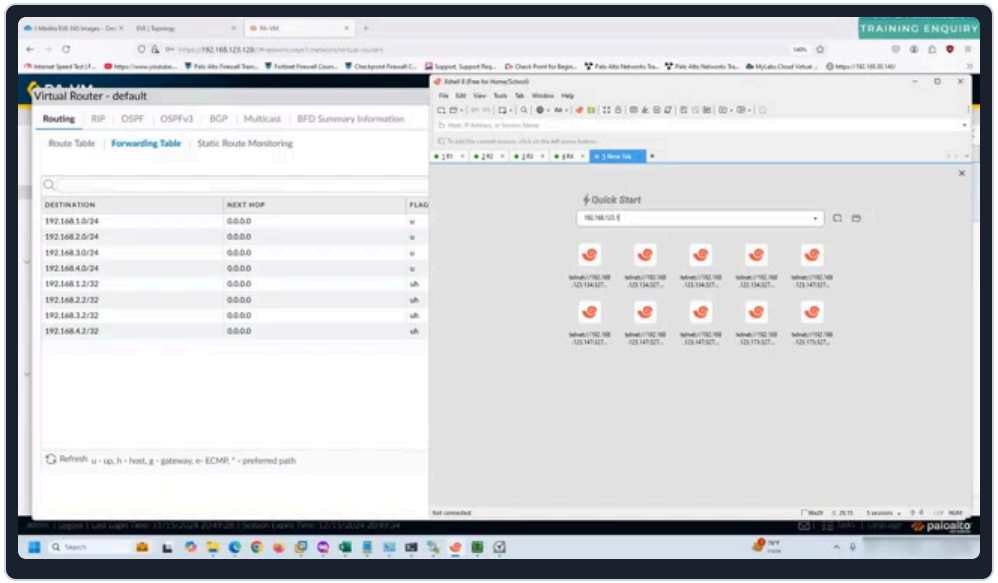


CORE CONCEPTS & KEY HIGHLIGHTS

- Logical partitioning of a single physical firewall into multiple independent virtual firewalls.
- Complete isolation of traffic, routing tables, security rules, NAT, and administrators.
- Ideal for Managed Service Providers (MSPs), large multi-division enterprises, and compliance segregation.

Virtual Systems Multi-Tenancy Architecture

Timestamp: 04:02:00

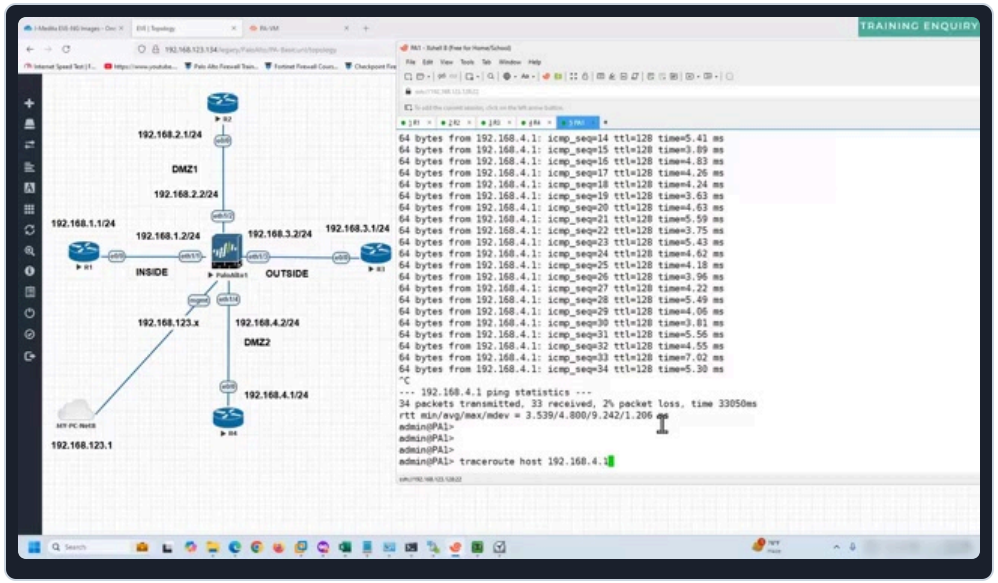


CORE CONCEPTS & KEY HIGHLIGHTS

- Each VSYS operates as a standalone security domain with its own zones, virtual routers, and policies.
- Shared physical hardware acceleration (SP3 engine, Single-Pass classification) across all virtual systems.
- Eliminates hardware sprawl while maintaining strict isolation between tenant environments.

VSYS Hardware Support, Platform Matrix & Licenses

Timestamp: 04:05:00

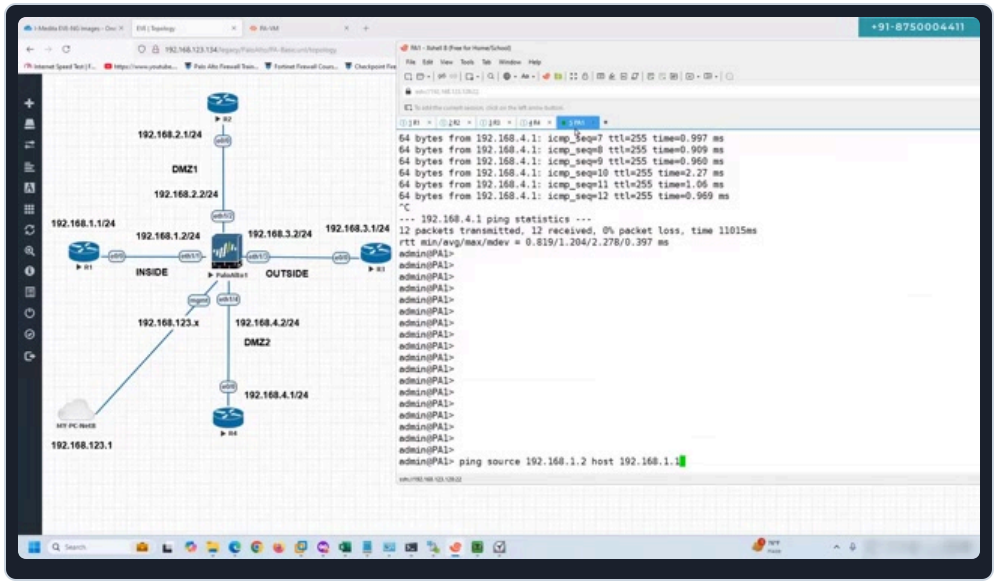


CORE CONCEPTS & KEY HIGHLIGHTS

- Supported on PA-3000, PA-3200, PA-3400, PA-5000, PA-5200, PA-5400, PA-7000 Series, and VM-300 / VM-Flex.
- Base VSYS included by default with optional multi-vsys expansion licenses (e.g. up to 225 on PA-7000).
- Note: Entry-level branch models (PA-220, PA-400) do not support multi-VSYS.

VSYS Partitioning: Independent Policies, NAT & Objects

Timestamp: 04:08:00

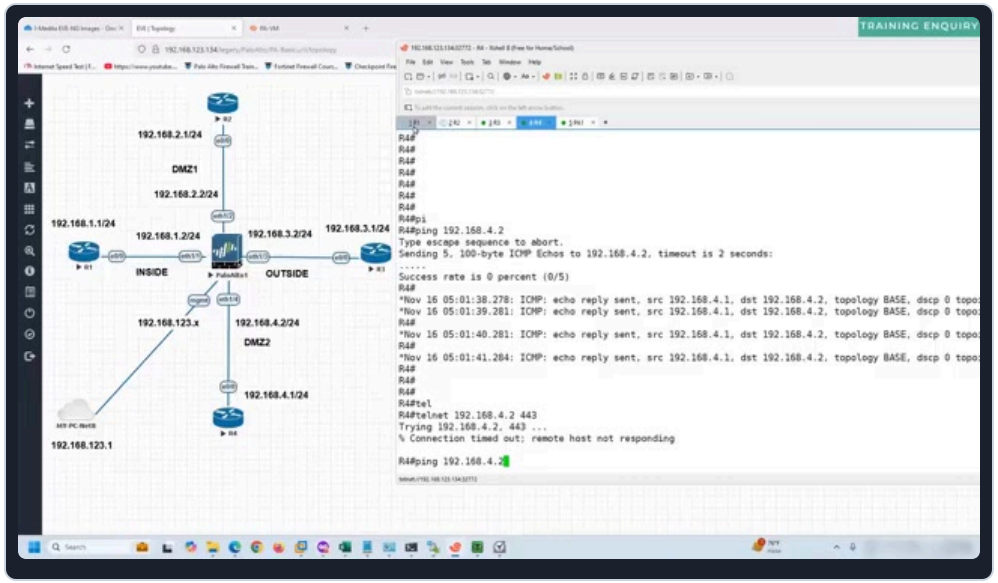


CORE CONCEPTS & KEY HIGHLIGHTS

- Independent Rulebase: Tenant A cannot view or modify Tenant B's Security or NAT policies.
- Dedicated Object Repositories: Custom address objects, service groups, and URL categories per VSYS.
- Global Shared Objects: Master administrator can publish global security profiles accessible across all VSYS.

Inter-VSYS Routing & Internal Packet Exchange

Timestamp: 04:12:00

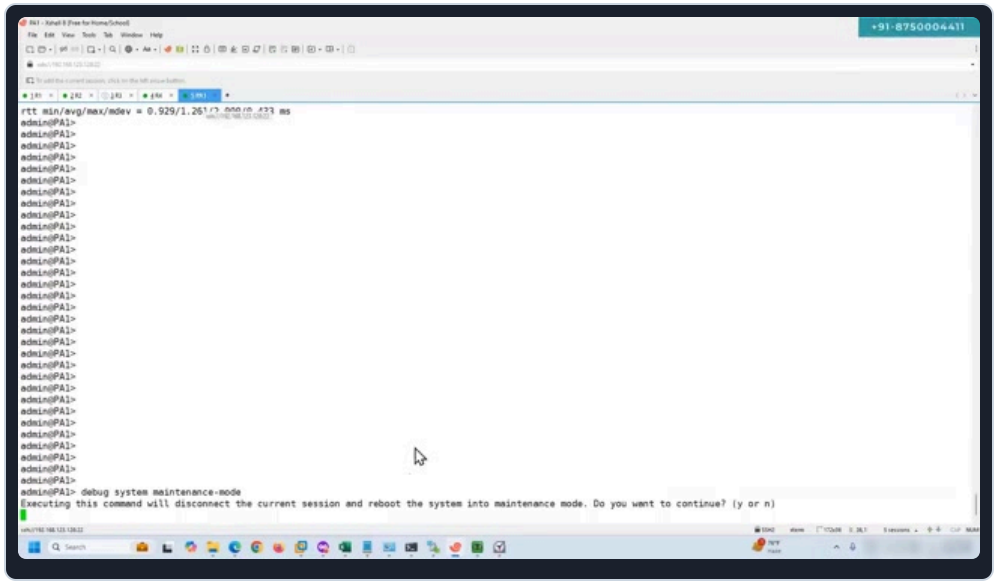


CORE CONCEPTS & KEY HIGHLIGHTS

- Routing traffic between distinct VSYS instances internally inside the firewall without physical loopback cables.
- Utilizes internal Virtual Routers or Shared Gateways for ultra-low latency, line-rate inter-tenant routing.
- Enforces Security Policies and Content-ID threat inspection on all inter-VSYS traffic flows.

Security Zones & Policy Rule Evaluation

Timestamp: 04:15:00

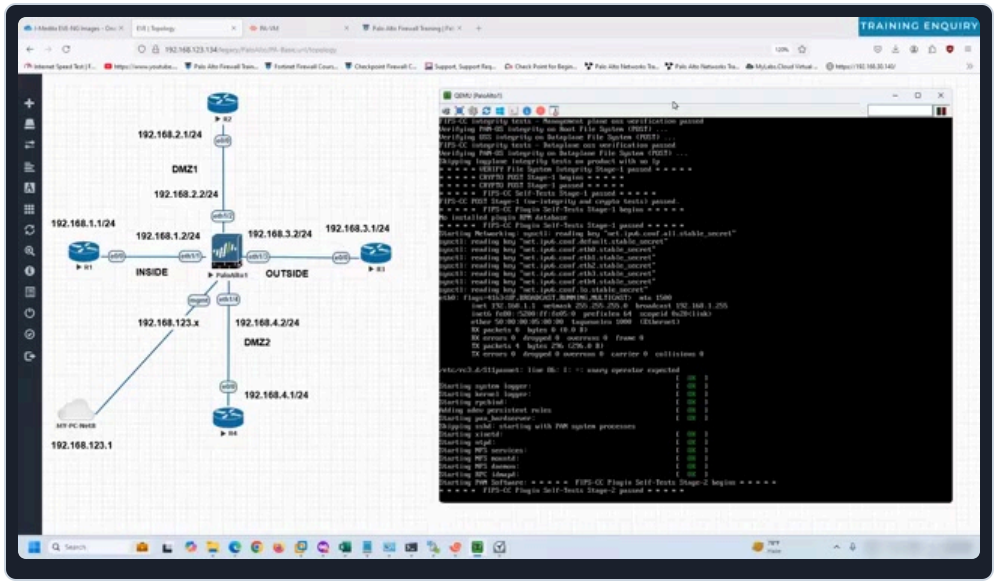


CORE CONCEPTS & KEY HIGHLIGHTS

- Zero-Trust zone model: Intra-zone allowed, Inter-zone denied by default.
- Top-to-bottom, first-match rule processing mechanics.
- Rule anatomy: Source Zone/IP, User, Dest Zone/IP, App-ID, Service, Action.

Shared Gateways & Centralized WAN Edge Architecture

Timestamp: 04:18:00

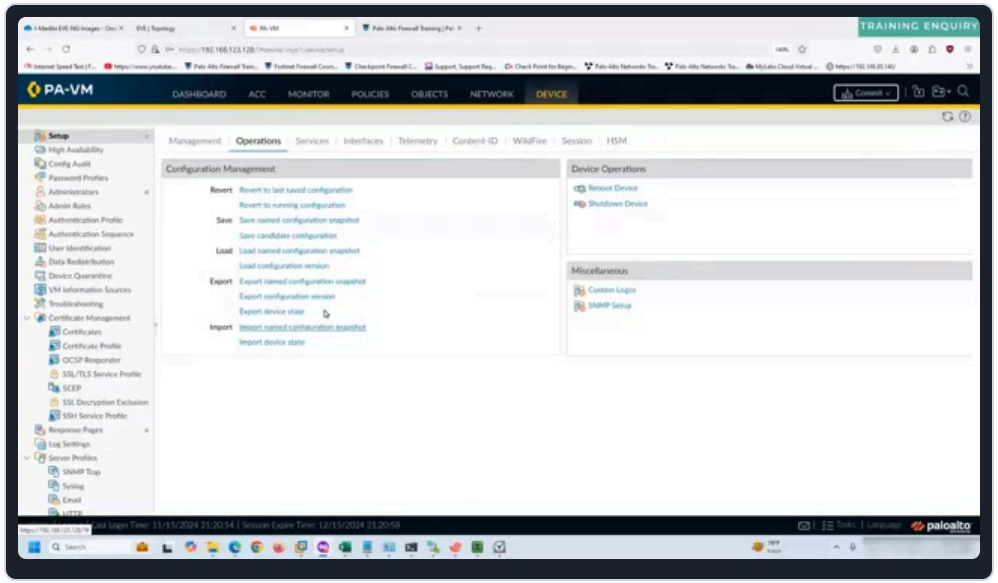


CORE CONCEPTS & KEY HIGHLIGHTS

- Shared Gateway allows multiple VSYS tenants to share a single public internet uplink and public IP pool.
- Centralized perimeter security enforcement while delegating internal departmental policies to individual VSYS.
- Simplifies WAN management and reduces public IP subnet consumption.

Interface & Zone Binding to Virtual Systems

Timestamp: 04:20:00

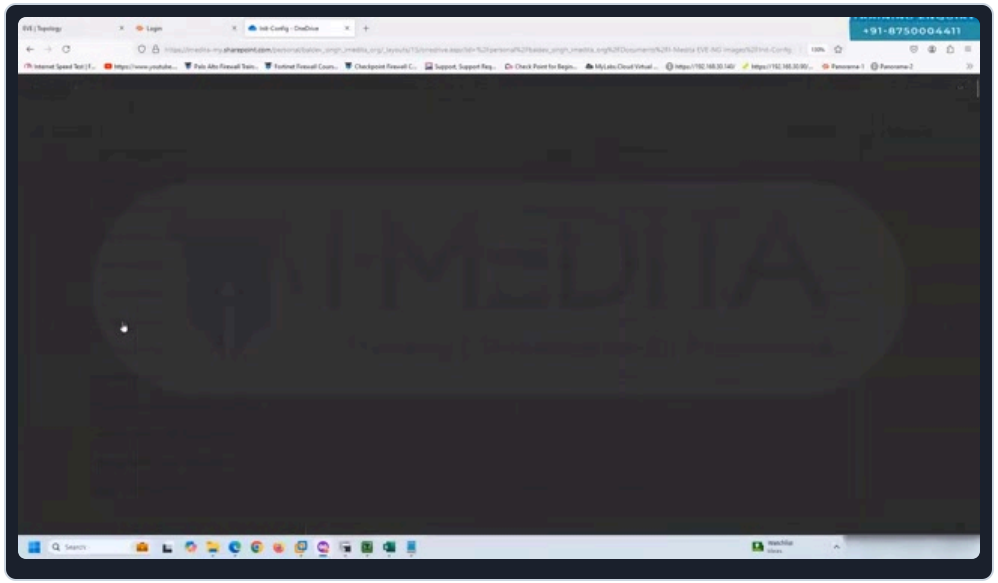


CORE CONCEPTS & KEY HIGHLIGHTS

- Binding physical interfaces, 802.1Q sub-interfaces, and VLAN interfaces to specific VSYS instances.
- Security zones are strictly scoped within their parent VSYS (e.g. `vsys1-Trust`, `vsys2-Trust`).
- VLAN tagging allows multiple tenants to share trunked physical switch links into separate VSYS.

Multi-Tenant Administrator Roles & Role-Based Access

Timestamp: 04:25:00

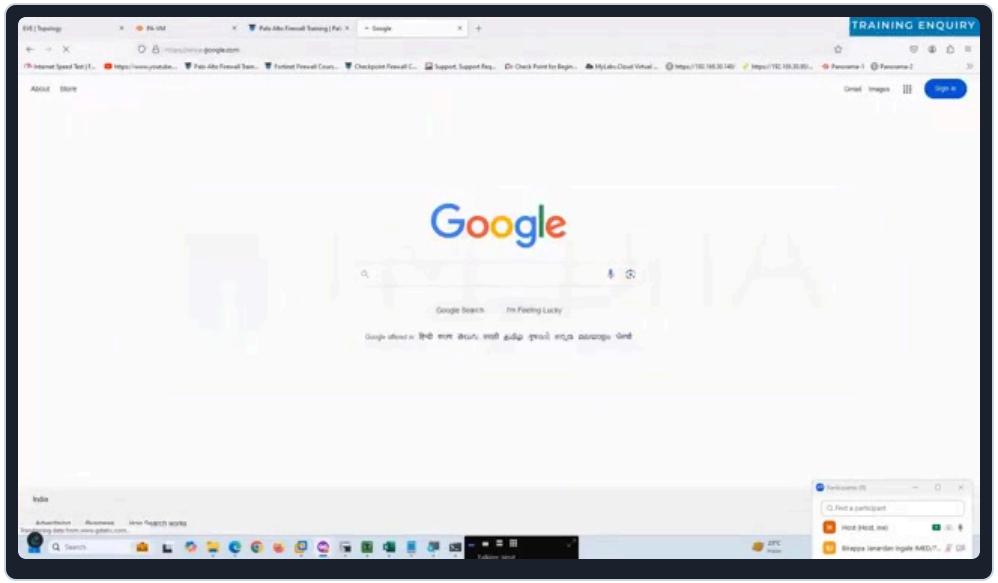


CORE CONCEPTS & KEY HIGHLIGHTS

- Superuser (Device Administrator): Full control over all VSYS, hardware settings, and global objects.
- VSYS Administrator: Restricted strictly to managing their designated virtual system(s).
- Tenant isolation prevents unauthorized changes or audit visibility into other departments or clients.

Resource Allocation: Session, Rule & NAT Limits per VSYS

Timestamp: 04:30:00

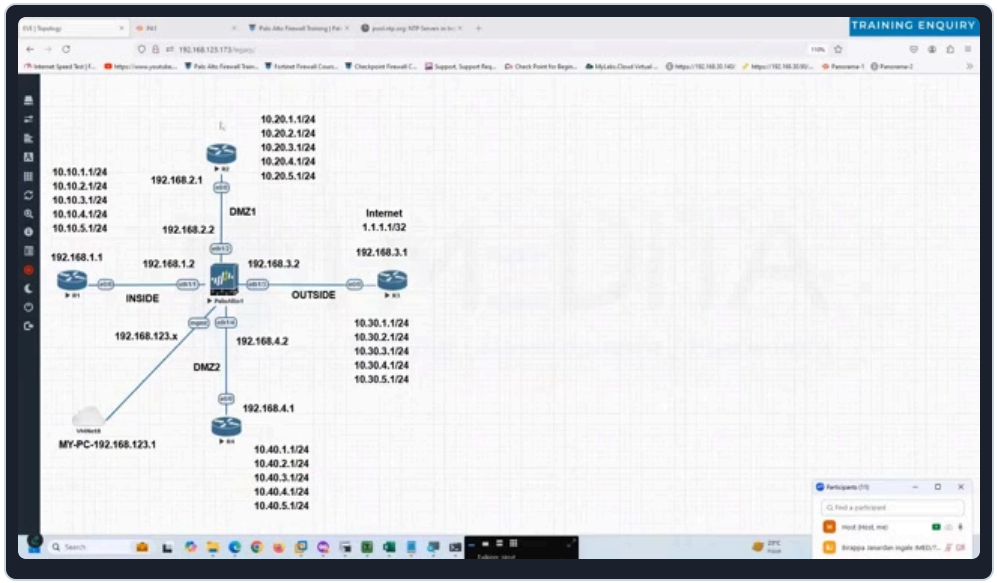


CORE CONCEPTS & KEY HIGHLIGHTS

- Resource Profiles prevent a single rogue tenant from exhausting firewall CPU or session memory (Noisy Neighbor).
- Enforceable limits per VSYS: Max concurrent sessions, Security Rules, NAT Rules, VPN Tunnels, and Decryption sessions.
- Dynamic real-time monitoring and alerting when a VSYS approaches its configured resource threshold.

Interface Configuration & Management Profiles

Timestamp: 04:40:00

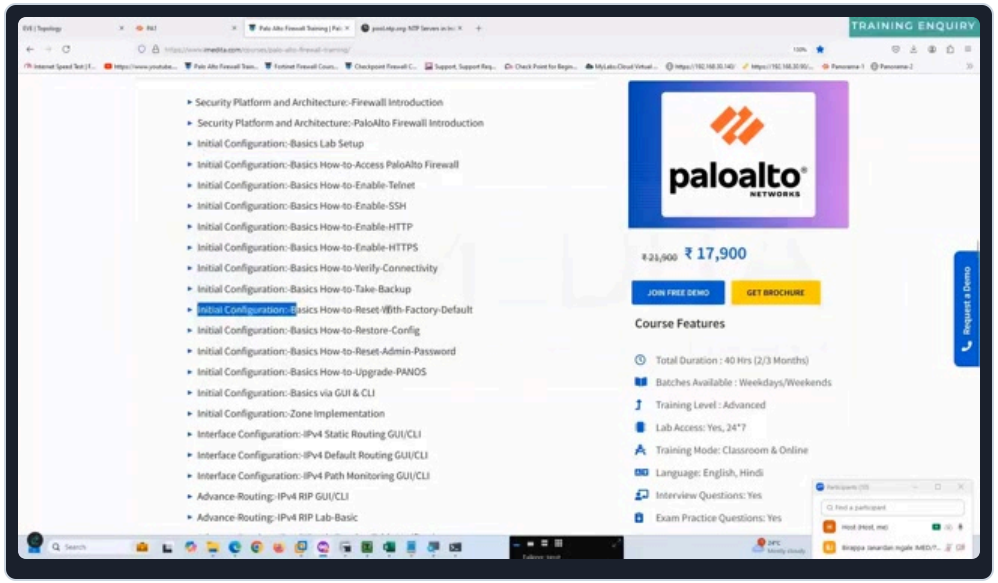


CORE CONCEPTS & KEY HIGHLIGHTS

- Assigning Layer 3 interfaces to Virtual Routers and Security Zones.
- Default block of ping/SSH on data plane interface IP addresses.
- Creating and applying Interface Management Profiles to permit ICMP ping.

Virtual Routers & Static Routing Configuration

Timestamp: 05:05:00

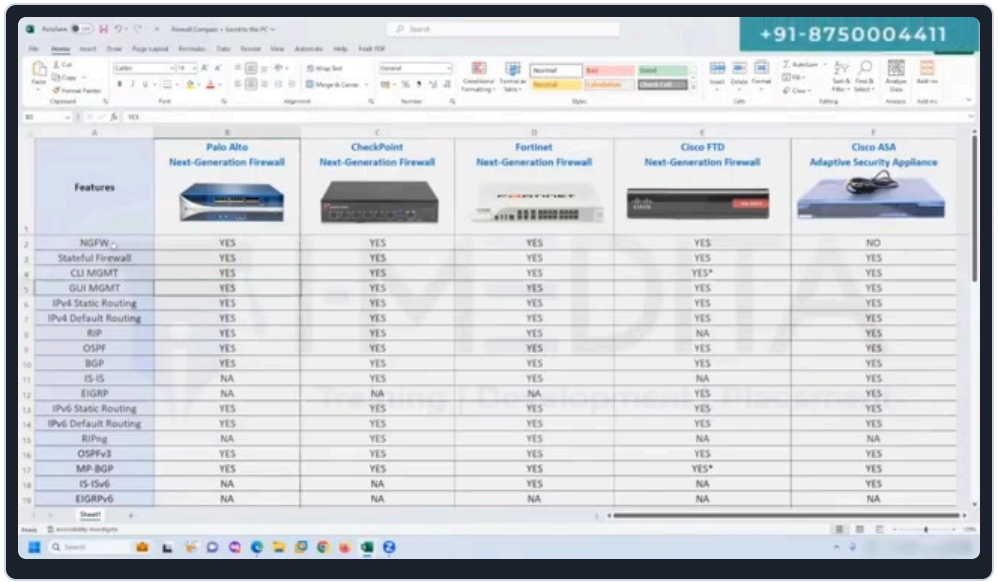


CORE CONCEPTS & KEY HIGHLIGHTS

- Decoupling routing tables using Virtual Routers (e.g. `default`).
- Configuring default route (0.0.0.0/0) pointing to upstream ISP gateway.
- Administrative distance, route metrics, and 802.1Q tagged sub-interfaces.

Path Monitoring & Dynamic WAN Failover (IP SLA)

Timestamp: 05:30:00



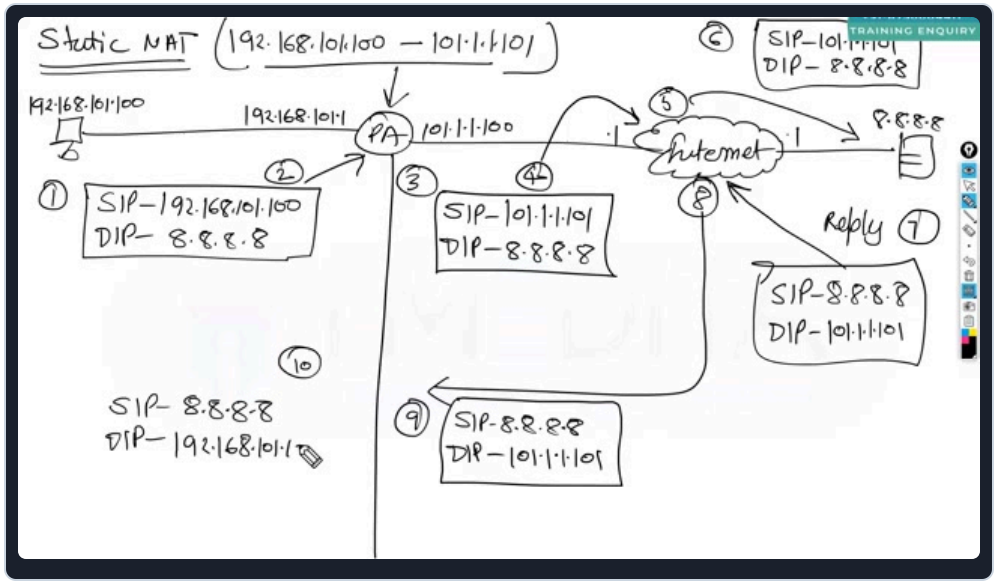
Features	Palo Alto Next-Generation Firewall	CheckPoint Next-Generation Firewall	Fortinet Next-Generation Firewall	Cisco FTD Next-Generation Firewall	Cisco ASA Adaptive Security Appliance
NGFW	YES	YES	YES	YES	NO
Stateful Firewall	YES	YES	YES	YES	YES
CLI MGMT	YES	YES	YES	YES*	YES
GUI MGMT	YES	YES	YES	YES	YES
IPv4 Static Routing	YES	YES	YES	YES	YES
IPv4 Default Routing	YES	YES	YES	YES	YES
RIP	YES	YES	YES	NA	YES
OSPF	YES	YES	YES	YES	YES
BGP	YES	YES	YES	YES	YES
IS-IS	NA	YES	YES	NA	YES
EIGRP	NA	NA	NA	YES	YES
IPv6 Static Routing	YES	YES	YES	YES	YES
IPv6 Default Routing	YES	YES	YES	YES	YES
RIPng	NA	YES	YES	NA	NA
OSPFv3	YES	YES	YES	YES	YES
MP-BGP	YES	YES	YES	YES*	YES
IS-ISv6	NA	NA	YES	NA	YES
EIGRPv6	NA	NA	NA	NA	NA

CORE CONCEPTS & KEY HIGHLIGHTS

- Attaching ICMP ping probes to next-hop gateway monitoring.
- Automatic route withdrawal and failover to backup ISP on line failure.
- Configuring probe intervals and failure thresholds.

Source NAT: Dynamic IP and Port (DIPP / PAT)

Timestamp: 06:00:00

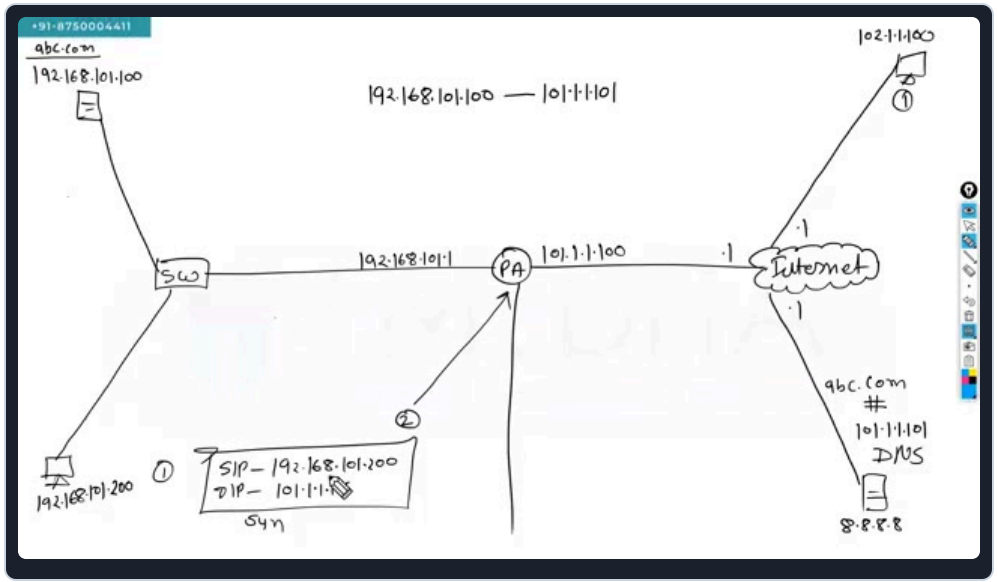


CORE CONCEPTS & KEY HIGHLIGHTS

- Translating private LAN subnets to public interface IP with port multiplexing.
- Pre-NAT Source/Destination vs Post-NAT interface translation.
- Up to 64,000 concurrent sessions per IP address.

Destination NAT: Inbound Port Forwarding

Timestamp: 06:30:00

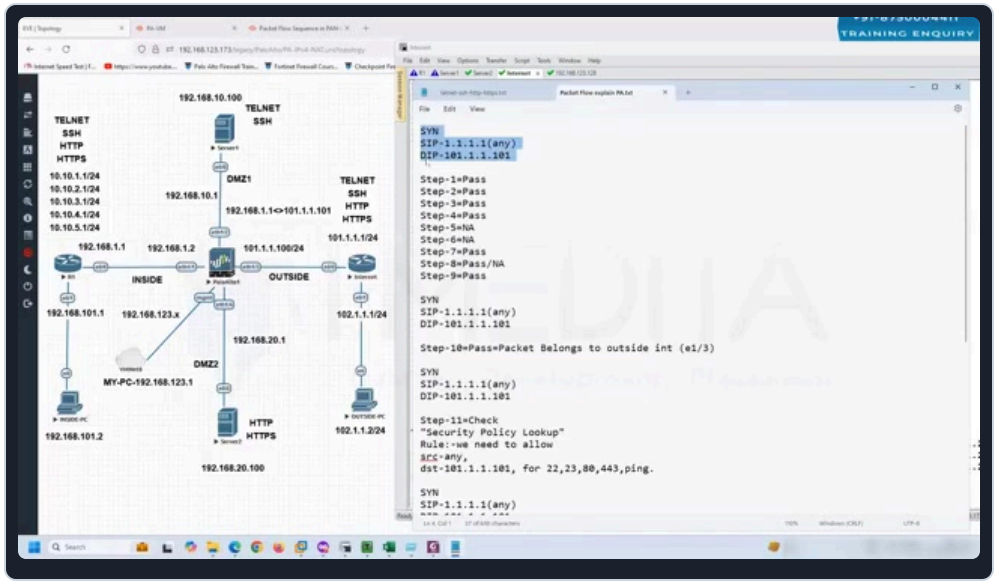


CORE CONCEPTS & KEY HIGHLIGHTS

- Publishing internal DMZ servers to public internet.
- Translating public destination IP:port to private server IP:port.
- Security policy configuration matching pre-NAT destination IP and post-NAT zone.

U-Turn NAT (Hairpinning) for Internal Access

Timestamp: 07:00:00

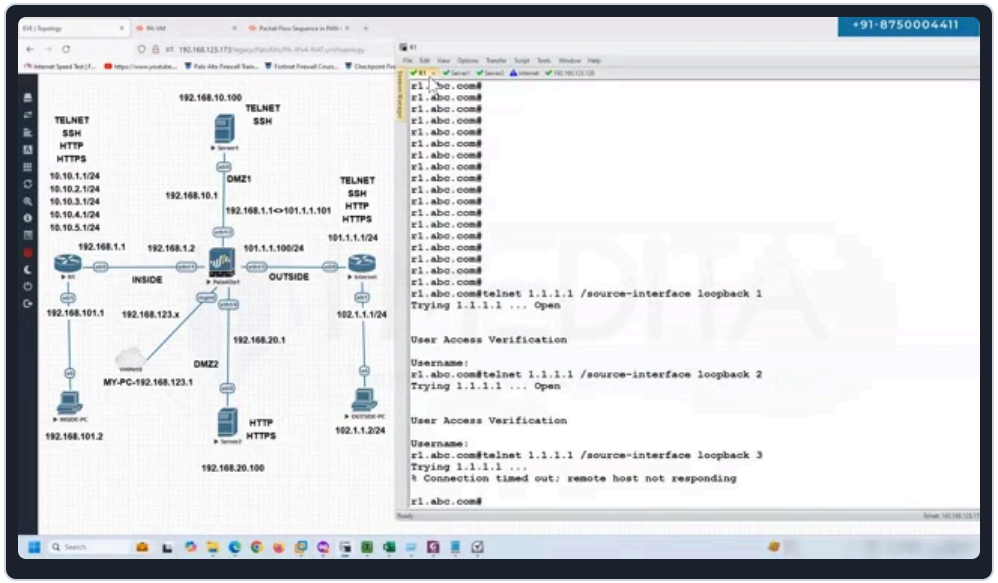


CORE CONCEPTS & KEY HIGHLIGHTS

- Allowing internal LAN users to access internal servers via public IP/DNS.
- Source NAT + Destination NAT dual-translation on the same physical interface.
- Preventing asymmetric routing drops inside the LAN.

App-ID™ Protocol Decoders & Heuristics

Timestamp: 07:30:00

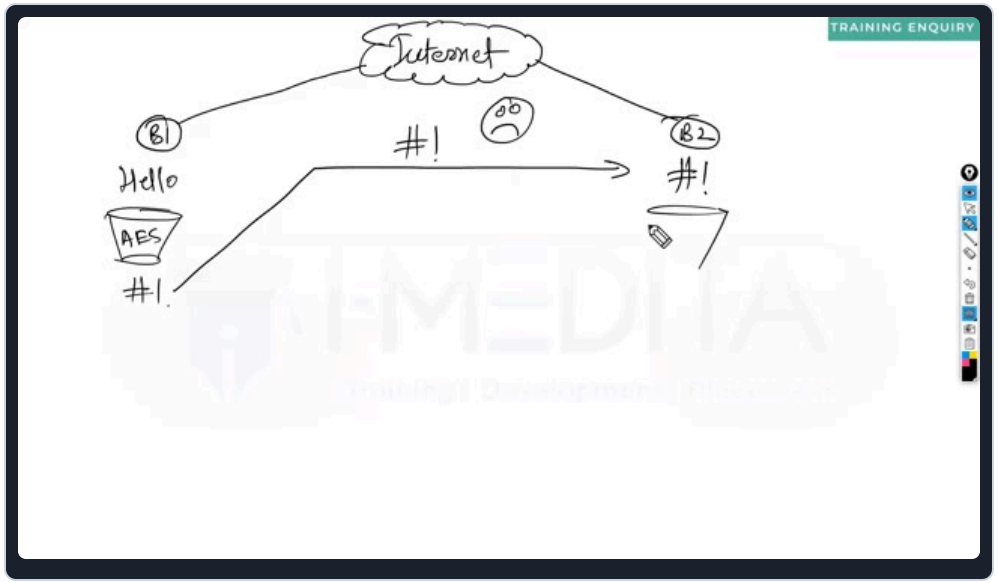


CORE CONCEPTS & KEY HIGHLIGHTS

- Payload inspection across initial packets to classify exact application.
- Handling application shifts (e.g. SSL -> Facebook -> Facebook-Chat).
- App-ID dependencies and parent/child application definitions.

Policy Optimizer & App-ID Migration

Timestamp: 08:00:00

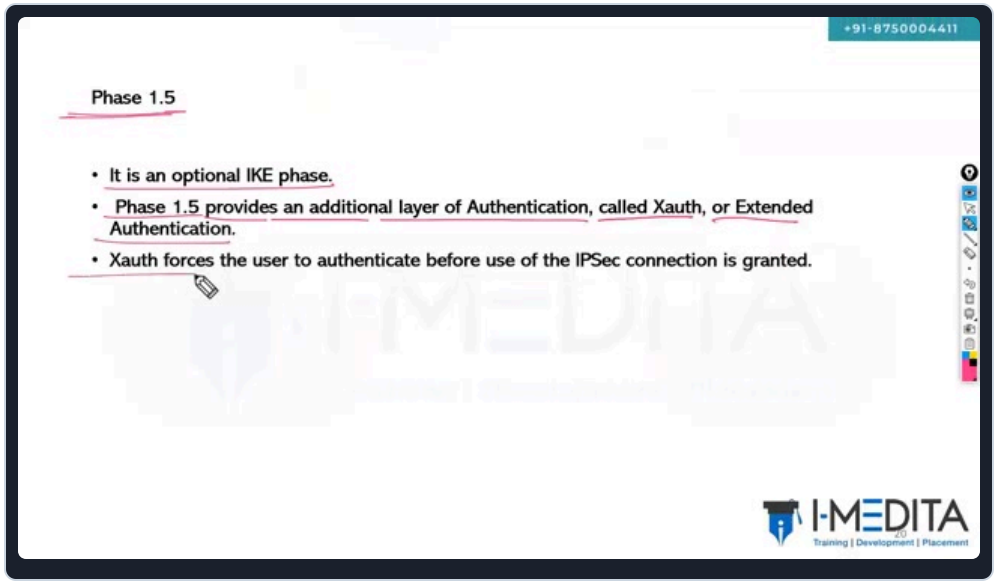


CORE CONCEPTS & KEY HIGHLIGHTS

- Analyzing legacy port-based rules (`any` application on port 80/443).
- Built-in Policy Optimizer monitoring observed App-IDs over time.
- 1-click migration to tight, least-privilege App-ID rules.

Content-ID™ Suite: Antivirus, Anti-Spyware & DNS Sinkholing

Timestamp: 08:30:00

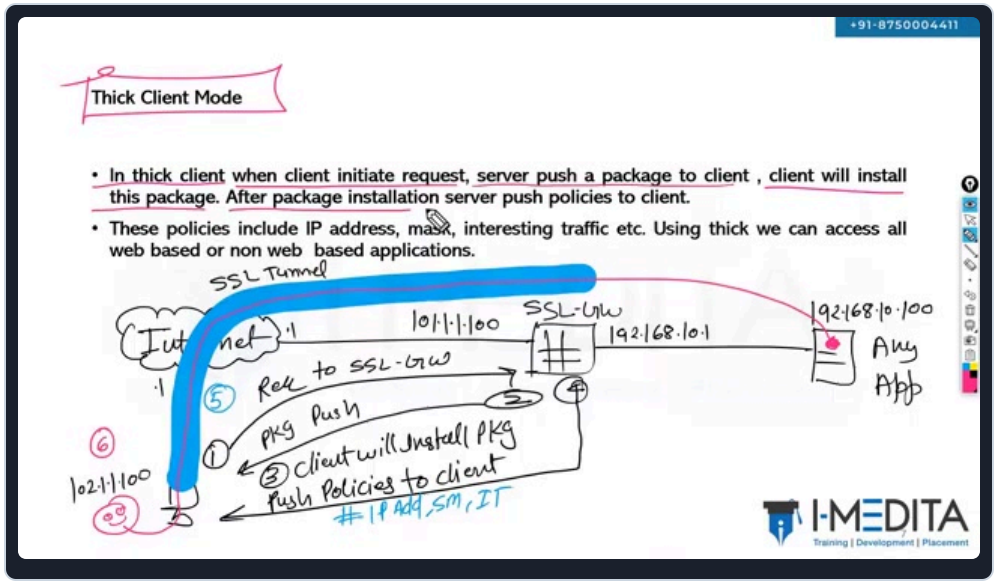


CORE CONCEPTS & KEY HIGHLIGHTS

- Stream-based Antivirus across HTTP, FTP, SMB, SMTP, and POP3.
- Anti-Spyware blocking C2 botnet communication.
- DNS Sinkholing diverting malicious domain queries to decoy IP for threat isolation.

Vulnerability Protection (IPS) & URL Filtering

Timestamp: 09:00:00



CORE CONCEPTS & KEY HIGHLIGHTS

- IPS signatures preventing buffer overflows, RCE, and OS exploits.
- URL Filtering across 80+ web categories.
- Action controls: `allow`, `alert`, `block`, `continue` (warning), and `override`.

User-ID™ Integration with Active Directory

Timestamp: 09:30:00

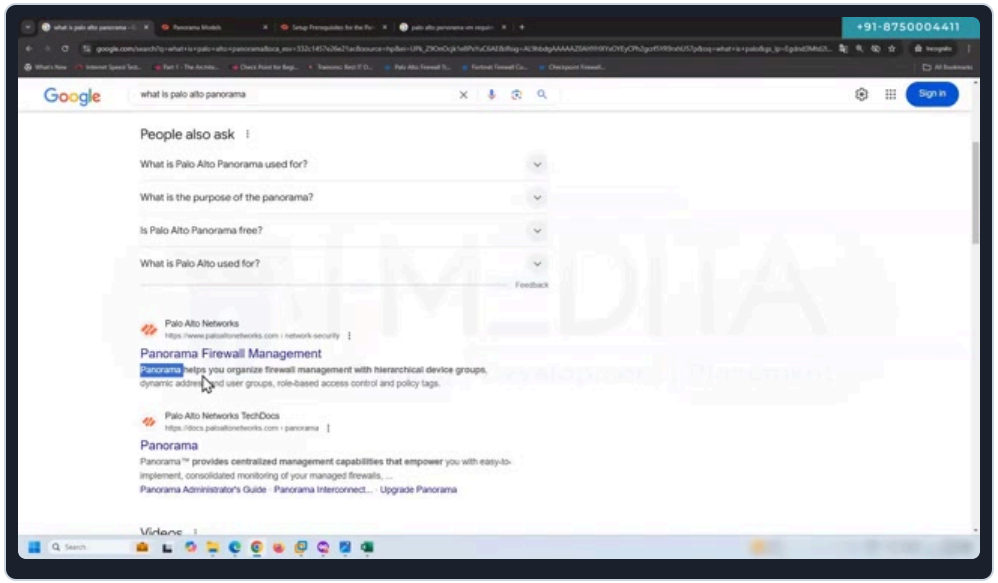


CORE CONCEPTS & KEY HIGHLIGHTS

- User-ID Agent querying Windows Event Logs (Events 4624/4625) via WMI/WinRM.
- Syslog listener parsing VPN, 802.1X, and wireless authentication.
- LDAP group mapping for role-based security rules.

SSL/TLS Decryption: Forward Proxy vs Inbound Inspection

Timestamp: 10:00:00

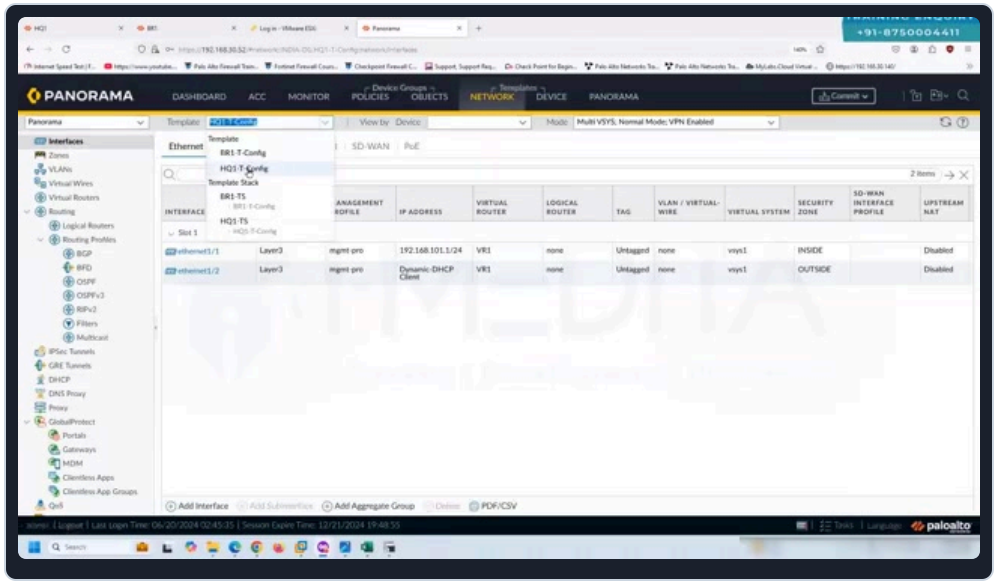


CORE CONCEPTS & KEY HIGHLIGHTS

- SSL Forward Proxy: Enterprise Subordinate CA inspecting outbound employee traffic.
- SSL Inbound Inspection: Protecting internal servers using private keys.
- Decryption exclusions for financial and healthcare compliance.

Site-to-Site IPsec VPN (IKEv1 / IKEv2)

Timestamp: 10:30:00

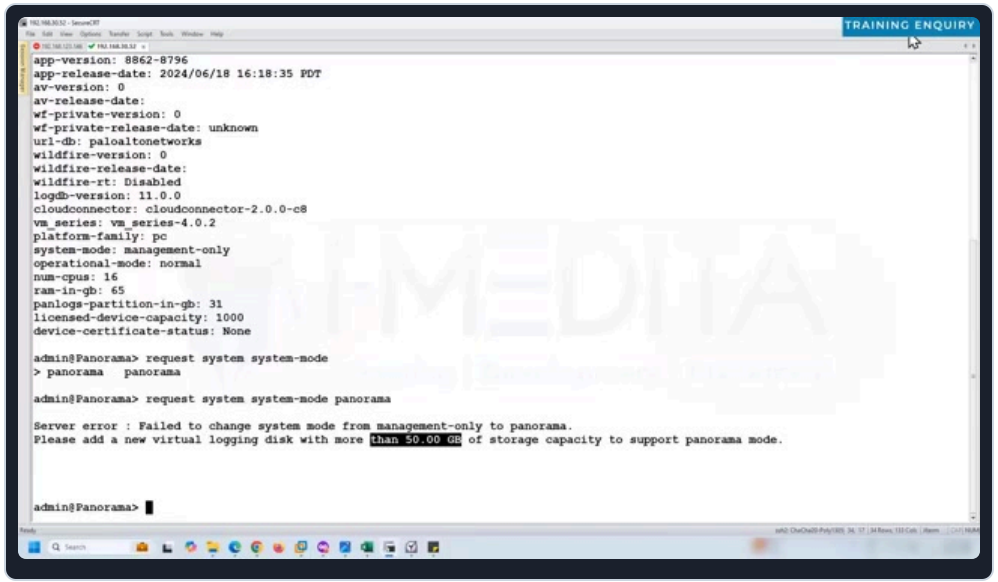


CORE CONCEPTS & KEY HIGHLIGHTS

- Route-based VPN architecture using virtual `tunnel.x` interfaces.
- IKE Phase 1 (ISAKMP management tunnel) & IKE Phase 2 (IPsec SA data encryption).
- DH Groups, AES-256-GCM encryption, and Perfect Forward Secrecy (PFS).

High Availability (Active/Passive) & Clustering Failover

Timestamp: 11:00:00



CORE CONCEPTS & KEY HIGHLIGHTS

- Active/Passive cluster state synchronization and sub-second failover.
- HA1 Control Link (Config & Heartbeats) and HA2 Data Link (Session State Table).
- Device Priority, Preemption, Link Monitoring, and Path Monitoring failover triggers.